

Diagnosing Mitigation Need: Benchmark Results for Controlled Shortages and Signal-Matched Interventions in Histopathology Patch Classification

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Abstract

A class can be underserved for several distinct reasons, and the count that marks it as rare does not say which. This report evaluates the controlled patch-classification benchmark specified in the companion protocol (Queisler, 2026a) on BRACS, a seven-class breast-lesion task represented by Virchow2 features. The design separates two axes that ordinary imbalance confounds: nominal patch support and the number of patients contributing those patches. Nominal deprivation produced material balanced-accuracy losses in eight of nine comparisons, with deficits of 1.5–13.8 percentage points; tail-group likelihood also worsened in eight comparisons, by up to 1.744 nats (natural-log units of predictive loss). Reducing patient breadth while holding the patch allocation fixed produced balanced-accuracy deficits of 2.6–3.0 percentage points across all three class assignments. Prevalence-weighted and class-balanced cross-entropy were the best methods for nominal damage. The largest probability-quality recovery estimate, 1.23 for class-balanced cross-entropy, means that the method surpassed the balanced reference by 23% of the original deficit after closing it; it does not imply that more than 100% of the lost probability mass was recovered. By contrast, no tested method repaired the damage caused by reduced patient breadth, and signal-matched weighting differed from unmatched weighting by only 0.06 percentage points of balanced accuracy. These results motivate mitigation that responds to dependence among patches from the same patient rather than to per-class counts alone.

Contents

1	Introduction	2
2	Allocation-Induced Damage	2
2.1	Realized signal profiles	3
2.2	Discrimination deficit	4
2.3	Probability-quality deficit	4
2.4	Where the damage falls	4
2.5	Natural anchor	5
3	Mitigation Recovery	5
3.1	The confirmatory five-family	5
3.2	Matched against unmatched signals	8
3.3	Recovery of probability quality	10
3.4	The wider method roster	13
4	Explaining Heterogeneity	19
5	Discussion	19
5.1	What the evidence supports	20
5.2	Which shortage a mitigation method must read	20
5.3	Signal against mechanism	21
5.4	Where recovery ceilings	21
5.5	Design requirements and open questions	22
5.6	Pending TCGA-UT analysis	22
6	Conclusion and Scope	22
	Appendices	23
A	Realized Conditions and Analysis Validity	23
B	Aggregate Damage Estimates	36
C	Per-Class and Tier Endpoints	36
D	Calibration Detail	40
E	Per-Partition Results	44
F	Computational Cost	44

1 Introduction

This report presents the results of the controlled patch-classification benchmark whose design, endpoints, materiality criteria, and inference procedure are specified in the companion protocol (Queisler, 2026a). The benchmark exists to inform the design of a new mitigation method, not to rank the existing ones. Its premise is that an underserved class may lack nominal patch support, independent evidence from patients or slides, learnable signal, or representational diversity, and that these shortages are not interchangeable. Existing methods are therefore treated as instruments: each reads one signal, and the contrast among them measures whether reading the right signal matters. The report is consequently read forward into Section 5, where the outcomes are converted into requirements a method would have to satisfy. Dataset provenance is documented separately (Queisler, 2026b), as is the signal taxonomy that locates each method within it (Queisler, 2026c).

The protocol poses three questions, restated here so that the results can be read without the companion document at hand:

RQ1. Which shortage does controlled allocation create, and what does it damage?

Under otherwise fixed conditions, how do controlled changes in training allocation affect tail-class discrimination and calibration, and which signal profile does each condition realize?

RQ2. Does signal-matched mitigation recover more of the damage? Which methods recover performance lost specifically to training imbalance, and does recovery improve when a method reads the shortage identified in RQ1?

RQ3. Which realized signal shortages explain variation in damage and mitigation headroom? Across controlled conditions, are allocation severity, independent-evidence shortage, support–difficulty alignment, and diversity shortage associated with damage magnitude and subsequent recovery?

Sections 2 and 3 answer the first two. Section 4 specifies the association models for the third. Statistical tests are restricted to the five weighting methods and to comparisons meeting the prespecified materiality and precision criteria; estimates for the wider method roster remain exploratory.

2 Allocation-Induced Damage

Four prerequisites must hold before the outcome measures are interpreted:

1. the moderate and severe allocations must reach their target ranges;
2. the concentrated and spread allocations must differ in patient breadth while retaining identical patch counts;
3. the pooled analysis must contain enough contributing patients for stable interval estimates; and
4. every planned method, initialization seed, and data partition must be present.

A failed allocation would invalidate the intended comparison, insufficient patient support would limit the corresponding estimates to summaries, and incomplete results would not be aggregated. Appendix A provides the full checks, tuning selections, and protocol deviations. The companion protocol describes the four deprived conditions and their two reference allocations (Queisler, 2026a).

All four prerequisites were met. Moderate and severe allocations achieved $\rho = 10.00$ and 100.38, respectively, within their target ranges. The spread allocation retained exactly 16,796 patches at $\rho = 1.000$ while increasing the contributing-patient count from 85 to the full eligible inventory of 103. Every class had sufficient patient support in the pooled analysis. All three partitions contained five initialization seeds for every method block and the complete set of 378 comparisons. The analysis therefore contains 12 comparison units: three class assignments crossed with four deprived conditions.

2.1 Realized signal profiles

Allocation severity is the manipulated variable, but it is not the explanatory one. Each controlled condition is characterized instead by the signal profile it realizes: achieved imbalance ratio ρ and normalized-entropy deficit $1 - H_{\text{norm}}$, nominal patch-support shortage, independent patient-support shortage, support-difficulty alignment, and representational-diversity shortage. These quantities are measured from the realized training sets rather than inferred from the target ratio. Each condition is compared with the reference that isolates its axis: concentrated balanced for the concentrated moderate and severe conditions, and balanced spread for concentrated balanced and severe spread.

Assignment	ρ	$1 - H_{\text{norm}}$	S_{nom}	S_{ind}	S_{div}	A	Dominant
<i>Concentrated balanced vs balanced spread</i>							
Native	1.00	0.000	-1.47	1.73	-1.32	0.20	Independent support
Aligned	1.00	0.000	-1.47	1.73	-1.32	0.20	Independent support
Reversed	1.00	0.000	-1.47	1.73	-1.31	0.20	Independent support
<i>Concentrated moderate vs concentrated balanced</i>							
Native	10.00	0.128	-0.35	-0.58	-0.79	0.36	Nominal support
Aligned	10.00	0.128	-0.17	-0.58	-0.35	-0.92	Difficulty
Reversed	10.00	0.128	-0.39	-0.58	-0.65	0.92	Nominal support
<i>Concentrated severe vs concentrated balanced</i>							
Native	100.38	0.353	0.76	-0.58	0.80	0.36	Ambiguous
Aligned	100.38	0.353	1.26	-0.58	1.28	-0.92	Difficulty
Reversed	100.38	0.353	0.65	-0.58	0.79	0.92	Ambiguous
<i>Severe spread vs balanced spread</i>							
Native	100.38	0.353	0.76	-0.58	1.00	0.36	Ambiguous
Aligned	100.38	0.353	1.26	-0.58	0.88	-0.92	Difficulty
Reversed	100.38	0.353	0.65	-0.58	0.99	0.92	Diversity

Table 1: BRACS signal profiles for the four deprived conditions. Each block names the deprived allocation and its reference; rows give the three class assignments. Shortage scores and alignment are standardized across the 12 comparison units. A dominant shortage is marked ambiguous when the two largest scores differ by less than 0.25 standard deviations.

The independent-support comparison holds 16,796 patches fixed but draws them from fewer patients. Across partitions, the affected classes retain about 51–66% of the patient support available in the spread reference, corresponding to a mean log-shortage of 0.409–0.673 nats. This is the largest patient-breadth contrast compatible with the contribution limits.

The three concentrated-balanced rows reuse one training allocation and differ only in the class assignment used for comparison; they are correlated views, not independent replications. Three severe comparisons are ambiguous because their two largest shortage scores are nearly tied: native severe, reversed severe, and native severe spread.

2.2 Discrimination deficit

The discrimination deficit D_{BA} is the balanced accuracy lost relative to the appropriate reference, reported in percentage points. The native assignment follows clinical class order; the difficulty-aligned assignment gives the fewest patches to the hardest classes, whereas the reversed assignment gives those classes the most patches. Recovery is analysed when the deficit reaches 1.44 percentage points and its paired 95% interval excludes zero.

Moderate imbalance produces deficits of 1.50–8.17 percentage points, concentrated severe imbalance 4.30–13.23, and severe spread 4.04–13.79. Concentrating the balanced allocation on fewer patients costs 2.64–2.95 percentage points despite identical patch counts. The only exception is difficulty-aligned moderate: its 1.50-point estimate clears the magnitude threshold, but its interval includes zero. Full estimates and intervals appear in Appendix Table 22.

2.3 Probability-quality deficit

Probability quality is measured by tail-group macro negative log likelihood. One *nat* is one natural-log unit of predictive loss, and lower values are better. Recovery is analysed when deprivation increases this loss by at least 0.148 nats and the paired 95% interval excludes zero.

Moderate imbalance increases loss by 0.186–0.754 nats, concentrated severe imbalance by 0.688–1.744, and severe spread by 0.201–1.127. Native moderate is the exception because its interval includes zero. The independent-support comparisons increase loss by 0.132–0.135 nats, just below the magnitude threshold, so they do not enter probability-quality recovery analysis. Full estimates, intervals, and secondary calibration measures appear in Appendix Table 23.

2.4 Where the damage falls

Aggregate deficits do not show which support tier absorbs the loss. Table 2 therefore retains only recall for the two difficulty-based assignments; full per-class endpoints appear in Appendix C. Tier labels in the balanced spread rows reflect class ordering rather than unequal support because that allocation is uniform.

Condition	Aligned recall (%)			Reversed recall (%)		
	Head	Body	Tail	Head	Body	Tail
Balanced spread	66.8	32.2	40.8	40.7	31.0	66.9
Moderate	76.4	37.6	18.3	46.1	19.7	45.9
Severe	74.7	41.0	11.8	48.5	15.3	35.5
Severe spread	73.8	40.8	13.2	51.2	13.3	31.9

Table 2: Tier recall (%) under aligned and reversed assignments.

Under the aligned assignment, tail recall falls from 40.8% under balanced spread to 18.3% at moderate and 11.8% at severe imbalance, while head recall remains above 73%. Under the reversed assignment, the body tier is weakest: recall falls from 31.0% under balanced spread to 15.3% at severe and 13.3% at severe spread. Difficulty therefore changes where damage appears; support rank alone does not identify the affected tier.

2.5 Natural anchor

The natural condition trains on all 217,399 eligible patches at the dataset’s observed prevalence. It is not size-matched and enters no estimand; it only places the controlled results beside the training distribution encountered without intervention.

Condition	Training patches	Balanced accuracy (%)
Natural	217,399	49.8
Concentrated balanced	16,796	49.2
Balanced spread	16,796	50.5–50.9

Table 3: Natural and balanced cross-entropy anchors.

Despite using nearly thirteen times as many patches, the natural condition reaches 49.8% balanced accuracy, compared with 49.2% for concentrated balanced and 50.5–50.9% across the balanced spread assignments. Additional patches at natural prevalence therefore buy less discrimination than distributing the controlled budget across more patients.

3 Mitigation Recovery

Recovery is defined as $R_M = [M(\text{method}) - M(\text{deprived CE})]/D_M$, the fraction of an established deficit that a method returns. Comparisons that do not meet the materiality and precision criteria in Sections 2.2 and 2.3 have no stable recovery fraction and are omitted.

3.1 The confirmatory five-family

The confirmatory family contains one method per signal, and all five apply a mean-one class weight to unmodified cross-entropy: prevalence-weighted cross-entropy, class-balanced cross-entropy, independent-support weighted cross-entropy, pilot-difficulty weighted cross-entropy, and semantic-scale weighted cross-entropy. Holding the intervention mechanism fixed in this way is what makes the comparison a contrast between signals rather than between loss surgeries of differing aggressiveness. Each method is tested against deprived cross-entropy by a paired partition-block permutation test. The Holm adjustment is applied jointly over the dataset’s confirmatory family, which holds 138 members — five methods in each of the 12 units on each of the two axes, plus a matched contrast for each non-ambiguous unit and axis — of which 108 sit on a gate-passing axis and are therefore testable.

Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status
BRACS	Native	Balanced	Prevalence	0.003	[-0.014, 0.029]	0.5728	1.0000	Tested
BRACS	Native	Balanced	Nominal support	0.005	[-0.013, 0.031]	0.3891	1.0000	Tested
BRACS	Native	Balanced	Independent support	0.013	[-0.008, 0.054]	0.1222	1.0000	Tested
BRACS	Native	Balanced	Difficulty	-0.057	[-0.258, 0.064]	0.3557	1.0000	Tested
BRACS	Native	Balanced	Diversity	0.021	[-0.226, 0.282]	0.7508	1.0000	Tested
BRACS	Aligned	Balanced	Prevalence	0.003	[-0.014, 0.031]	0.5728	1.0000	Tested
BRACS	Aligned	Balanced	Nominal support	0.005	[-0.012, 0.035]	0.3891	1.0000	Tested
BRACS	Aligned	Balanced	Independent support	0.013	[-0.009, 0.055]	0.1222	1.0000	Tested
BRACS	Aligned	Balanced	Difficulty	-0.058	[-0.267, 0.067]	0.3557	1.0000	Tested
BRACS	Aligned	Balanced	Diversity	0.021	[-0.248, 0.286]	0.7508	1.0000	Tested
BRACS	Reversed	Balanced	Prevalence	0.003	[-0.020, 0.040]	0.5728	1.0000	Tested
BRACS	Reversed	Balanced	Nominal support	0.005	[-0.018, 0.041]	0.3891	1.0000	Tested
BRACS	Reversed	Balanced	Independent support	0.015	[-0.010, 0.079]	0.1222	1.0000	Tested
BRACS	Reversed	Balanced	Difficulty	-0.064	[-0.366, 0.080]	0.3557	1.0000	Tested
BRACS	Reversed	Balanced	Diversity	0.024	[-0.326, 0.340]	0.7508	1.0000	Tested
BRACS	Native	Moderate	Prevalence	0.815	[0.419, 1.237]	0.0002	0.0130	Tested
BRACS	Native	Moderate	Nominal support	0.892	[0.549, 1.307]	<0.0001	0.0022	Tested
BRACS	Native	Moderate	Independent support	0.197	[-0.190, 0.557]	0.1928	1.0000	Tested
BRACS	Native	Moderate	Difficulty	0.121	[-0.263, 0.453]	0.4373	1.0000	Tested
BRACS	Native	Moderate	Diversity	0.142	[-0.415, 0.622]	0.2702	1.0000	Tested
BRACS	Reversed	Moderate	Prevalence	0.840	[0.720, 0.962]	<0.0001	0.0026	Tested
BRACS	Reversed	Moderate	Nominal support	0.833	[0.696, 0.969]	<0.0001	0.0011	Tested
BRACS	Reversed	Moderate	Independent support	0.051	[-0.144, 0.202]	0.5917	1.0000	Tested
BRACS	Reversed	Moderate	Difficulty	-0.078	[-0.244, 0.059]	0.2205	1.0000	Tested
BRACS	Reversed	Moderate	Diversity	-0.064	[-0.279, 0.126]	0.2047	1.0000	Tested
BRACS	Native	Severe	Prevalence	0.768	[0.441, 1.030]	0.0025	0.1214	Tested
BRACS	Native	Severe	Nominal support	0.637	[0.353, 0.880]	0.0028	0.1297	Tested
BRACS	Native	Severe	Independent support	0.106	[-0.149, 0.323]	0.2974	1.0000	Tested
BRACS	Native	Severe	Difficulty	-0.273	[-0.637, -0.070]	0.0012	0.0603	Tested
BRACS	Native	Severe	Diversity	-0.165	[-0.548, 0.049]	0.0280	1.0000	Tested
BRACS	Aligned	Severe	Prevalence	0.654	[0.330, 1.061]	0.0073	0.3344	Tested
BRACS	Aligned	Severe	Nominal support	0.426	[0.002, 0.767]	0.0614	1.0000	Tested
BRACS	Aligned	Severe	Independent support	0.024	[-0.193, 0.214]	0.7576	1.0000	Tested
BRACS	Aligned	Severe	Difficulty	0.164	[-0.071, 0.437]	0.0361	1.0000	Tested
BRACS	Aligned	Severe	Diversity	-0.059	[-0.524, 0.237]	0.5979	1.0000	Tested
BRACS	Reversed	Severe	Prevalence	0.794	[0.691, 0.889]	<0.0001	0.0006	Tested
BRACS	Reversed	Severe	Nominal support	0.736	[0.619, 0.856]	<0.0001	0.0006	Tested

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Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status
BRACS	Reversed	Severe	Independent support	0.111	[0.046, 0.184]	0.0010	0.0505	Tested
BRACS	Reversed	Severe	Difficulty	-0.059	[-0.120, 0.002]	0.0185	0.8122	Tested
BRACS	Reversed	Severe	Diversity	-0.010	[-0.092, 0.064]	0.7536	1.0000	Tested
BRACS	Native	Severe (spread)	Prevalence	0.867	[0.706, 1.050]	<0.0001	0.0016	Tested
BRACS	Native	Severe (spread)	Nominal support	0.827	[0.673, 1.007]	<0.0001	0.0006	Tested
BRACS	Native	Severe (spread)	Independent support	0.111	[-0.033, 0.226]	0.0411	1.0000	Tested
BRACS	Native	Severe (spread)	Difficulty	0.021	[-0.171, 0.190]	0.9236	1.0000	Tested
BRACS	Native	Severe (spread)	Diversity	-0.026	[-0.151, 0.074]	0.5866	1.0000	Tested
BRACS	Aligned	Severe (spread)	Prevalence	0.478	[0.005, 0.864]	0.0517	1.0000	Tested
BRACS	Aligned	Severe (spread)	Nominal support	0.467	[0.089, 0.785]	0.0297	1.0000	Tested
BRACS	Aligned	Severe (spread)	Independent support	-0.139	[-0.480, 0.060]	0.0738	1.0000	Tested
BRACS	Aligned	Severe (spread)	Difficulty	-0.149	[-0.853, 0.210]	0.1362	1.0000	Tested
BRACS	Aligned	Severe (spread)	Diversity	-0.149	[-0.638, 0.114]	0.0831	1.0000	Tested
BRACS	Reversed	Severe (spread)	Prevalence	0.823	[0.713, 0.939]	<0.0001	0.0006	Tested
BRACS	Reversed	Severe (spread)	Nominal support	0.790	[0.685, 0.903]	<0.0001	0.0006	Tested
BRACS	Reversed	Severe (spread)	Independent support	0.051	[-0.006, 0.113]	0.0204	0.8772	Tested
BRACS	Reversed	Severe (spread)	Difficulty	-0.008	[-0.091, 0.059]	0.6915	1.0000	Tested
BRACS	Reversed	Severe (spread)	Diversity	0.030	[-0.037, 0.091]	0.1580	1.0000	Tested

Table 5: Confirmatory recovery R_M for the five-member method family on discrimination-gate-passing units, with paired 95% intervals, permutation p -values, and Holm-adjusted p -values within the dataset. Signal labels name the family member: prevalence is prevalence-weighted cross-entropy, nominal support is class-balanced cross-entropy, and the remaining three are the independent-support, pilot-difficulty, and semantic-scale weighted variants.

The family separates into the same two groups in every unit where anything recovers at all. Prevalence-weighted and class-balanced cross-entropy return between 0.426 and 0.892 of the discrimination deficit across the eight nominal-axis units, and ten of their sixteen tests survive Holm adjustment. The independent-support, difficulty, and diversity members return between -0.149 and 0.197 on those same units, and not one of their 24 tests survives. The closest case is independent-support weighting under the reversed severe condition, at 0.111 with an interval of $[0.046, 0.184]$ and an adjusted p of 0.0505 ; on the prespecified rule it does not survive, and it is the only cell in which that method’s interval excludes zero on the discrimination axis.

The independent-support axis is where the family behaves differently, and the difference is this report’s central negative result. In all three balanced units every one of the five members recovers essentially nothing: point estimates from -0.064 to 0.024 , every interval covering zero, and every adjusted p equal to 1. The largest point estimate belongs to semantic-scale weighting, at 0.021 to 0.024 ; independent-support weighted cross-entropy, the member built to read exactly this shortage, returns 0.013 to 0.015 . A gate opened on a deficit of roughly three balanced-accuracy points, and the confirmatory family left all of it standing.

Two features of the nominal-axis results qualify the picture there. Holm adjustment bites hardest where recovery intervals are wide: the native severe and difficulty-aligned severe conditions carry prevalence-weighted point estimates of 0.768 and 0.654 and still fail adjustment, at $p = 0.121$ and $p = 0.334$. And the two prevalence-corner members do not separate here the way they can when an effective-number transform saturates; class-balanced cross-entropy leads in three units and prevalence weighting in five, with differences typically inside their intervals.

3.2 Matched against unmatched signals

The claim RQ2 actually tests is not that some method recovers damage but that the method reading the shortage a condition presents recovers more of it than a method reading a different shortage. Each gate-passing unit is therefore split into a matched arm, containing the family member whose signal corresponds to the dominant shortage from Table 1, and an unmatched arm containing the remainder. Ambiguous units contribute descriptively and carry no confirmatory weight.

Dataset	Assignment	Severity	Gate	Dominant	Contrast	95% CI (bootstrap)	Holm p	Best recoverer	Agrees
BRACS	Native	Balanced	Discrimination	Independent support	0.0006	[-0.0013, 0.0022]	1.0000	Diversity	No
BRACS	Aligned	Balanced	Discrimination	Independent support	0.0006	[-0.0013, 0.0022]	1.0000	Diversity	No
BRACS	Reversed	Balanced	Discrimination	Independent support	0.0006	[-0.0013, 0.0022]	1.0000	Diversity	No
BRACS	Native	Moderate	Discrimination	Nominal support	0.0234	[0.0096, 0.0375]	0.0367	Nominal support	Yes
BRACS	Reversed	Moderate	Discrimination	Nominal support	0.0708	[0.0525, 0.0915]	0.0006	Prevalence	Yes
BRACS	Native	Severe	Discrimination	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe	Discrimination	Difficulty	-0.0042	[-0.0163, 0.0086]	1.0000	Prevalence	No
BRACS	Reversed	Severe	Discrimination	Ambiguous	—	—	—	Prevalence	—
BRACS	Native	Severe (spread)	Discrimination	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe (spread)	Discrimination	Difficulty	-0.0127	[-0.0242, 0.0005]	0.4554	Prevalence	No
BRACS	Reversed	Severe (spread)	Discrimination	Diversity	-0.0529	[-0.0648, -0.0418]	0.0006	Prevalence	No
BRACS	Aligned	Moderate	Calibration	Difficulty	-0.5566	[-1.4624, 0.0712]	0.0004	Nominal support	No
BRACS	Reversed	Moderate	Calibration	Nominal support	1.3341	[1.0771, 1.5953]	0.0004	Prevalence	Yes
BRACS	Native	Severe	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe	Calibration	Difficulty	-0.4862	[-0.8830, -0.1232]	0.0004	Prevalence	No
BRACS	Reversed	Severe	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Native	Severe (spread)	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe (spread)	Calibration	Difficulty	-0.3229	[-0.5781, -0.1043]	0.0004	Nominal support	No
BRACS	Reversed	Severe (spread)	Calibration	Diversity	-0.7637	[-1.0321, -0.5357]	0.0004	Prevalence	No

Table 7: Direct contrast between the signal-matched arm and the unmatched arm on gate-passing units, with paired 95% intervals and Holm-adjusted p -values. Ambiguous units carry no matched arm and are descriptive only. “Best recoverer” names the family member with the largest recovery in that unit, and “Agrees” records whether it belongs to the matched arm.

The independent-support comparison directly tests its matched method on a condition that lacks patient support, and the contrast is flat. All three balanced units differ by 0.06 percentage points of balanced accuracy, with an interval of $[-0.13, 0.22]$ percentage points and an adjusted p of 1. The matching hypothesis is unsupported on this axis: neither arm contains a method that moves the endpoint.

Elsewhere the contrast is signed by which shortage the unit’s profile makes dominant, and the sign is predicted entirely by whether that shortage is nominal support. In the two units whose dominant shortage is nominal support the contrast is positive and survives adjustment: 0.0234 at native moderate and 0.0708 at reversed moderate. In every unit whose dominant shortage is difficulty or diversity the contrast is negative, reaching -0.0529 at reversed severe (spread) with an adjusted p of 0.0006, because the matched arm there holds exactly one member of the group that recovers nothing while the unmatched arm retains both members that work. The agreement column makes the same point without the arithmetic of the contrast: in every unit where any family member recovered a material fraction, that member was prevalence-weighted or class-balanced cross-entropy, so the dominant shortage and the best recoverer agree only where the dominant shortage is nominal support.

The matched- β diagnostic in Table 8 rules out the most obvious alternative explanation for the independent-support member’s failure. If it failed because its tuned weight strength was too timid rather than because its signal is uninformative, refitting it at the strength selected for class-balanced cross-entropy should close part of the gap.

Dataset	Assignment	Severity	CE	Independent	Independent at matched β	Nominal
BRACS	Native	Balanced	0.492	0.493	0.484	0.492
BRACS	Aligned	Balanced	0.492	0.493	0.484	0.492
BRACS	Reversed	Balanced	0.492	0.493	0.484	0.492
BRACS	Native	Moderate	0.467	0.479	0.476	0.495
BRACS	Aligned	Moderate	0.460	0.456	0.457	0.477
BRACS	Reversed	Moderate	0.423	0.434	0.431	0.486
BRACS	Native	Severe	0.447	0.456	0.454	0.479
BRACS	Aligned	Severe	0.429	0.430	0.433	0.461
BRACS	Reversed	Severe	0.382	0.398	0.385	0.473
BRACS	Native	Severe (spread)	0.432	0.447	0.451	0.503
BRACS	Aligned	Severe (spread)	0.431	0.427	0.425	0.478
BRACS	Reversed	Severe (spread)	0.375	0.378	0.379	0.491

Table 8: Matched- β diagnostic: case-macro balanced accuracy of independent-support weighted cross-entropy refitted at the effective weight strength selected for class-balanced cross-entropy, beside the deprived cross-entropy reference and the two anchors it separates. Descriptive, and not part of the Holm family.

It does not. Across all 12 units the refit moves balanced accuracy by at most 0.013 from the independent-support arm, and the two largest movements are downward, including on the balanced units, where the refit falls from 0.493 to 0.484. The gap to class-balanced cross-entropy meanwhile remains as large as 0.112, under the reversed severe (spread) condition. The failure of the non-prevalence signals is a property of what those weights encode, not of how hard they were applied.

3.3 Recovery of probability quality

A method may return discrimination while leaving probabilities no better calibrated, and a mitigation method that must serve both would need to be designed for that trade-off explicitly. Recovery on the tail-group negative log likelihood axis is therefore reported separately over the eight units whose calibration gate opened, which are not the same eight as on the discrimination axis.

Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status	Discrimination R_M
BRACS	Aligned	Moderate	Prevalence	1.197	[0.999, 1.537]	<0.0001	0.0004	Tested	—
BRACS	Aligned	Moderate	Nominal support	1.227	[1.006, 1.708]	<0.0001	0.0004	Tested	—
BRACS	Aligned	Moderate	Independent support	-0.838	[-2.759, -0.105]	0.0002	0.0013	Tested	—
BRACS	Aligned	Moderate	Difficulty	-1.352	[-6.467, 0.395]	<0.0001	0.0004	Tested	—
BRACS	Aligned	Moderate	Diversity	-1.125	[-2.574, -0.481]	<0.0001	0.0004	Tested	—
BRACS	Reversed	Moderate	Prevalence	0.887	[0.441, 0.997]	<0.0001	0.0004	Tested	0.840
BRACS	Reversed	Moderate	Nominal support	0.873	[0.353, 0.977]	<0.0001	0.0004	Tested	0.833
BRACS	Reversed	Moderate	Independent support	-0.274	[-4.425, 0.494]	0.0033	0.0098	Tested	0.051
BRACS	Reversed	Moderate	Difficulty	-1.095	[-6.533, -0.014]	<0.0001	0.0004	Tested	-0.078
BRACS	Reversed	Moderate	Diversity	-1.303	[-8.308, 0.186]	<0.0001	0.0004	Tested	-0.064
BRACS	Native	Severe	Prevalence	0.911	[0.743, 1.008]	<0.0001	0.0004	Tested	0.768
BRACS	Native	Severe	Nominal support	0.682	[0.177, 0.885]	0.0018	0.0070	Tested	0.637
BRACS	Native	Severe	Independent support	-1.287	[-3.215, -0.212]	0.0003	0.0015	Tested	0.106
BRACS	Native	Severe	Difficulty	-2.168	[-5.847, -0.879]	<0.0001	0.0004	Tested	-0.273
BRACS	Native	Severe	Diversity	-1.753	[-4.845, -0.655]	<0.0001	0.0004	Tested	-0.165
BRACS	Aligned	Severe	Prevalence	1.137	[1.043, 1.261]	<0.0001	0.0004	Tested	0.654
BRACS	Aligned	Severe	Nominal support	0.539	[0.069, 0.913]	<0.0001	0.0004	Tested	0.426
BRACS	Aligned	Severe	Independent support	-2.509	[-3.840, -1.587]	<0.0001	0.0004	Tested	0.024
BRACS	Aligned	Severe	Difficulty	-1.768	[-3.203, -0.779]	<0.0001	0.0004	Tested	0.164
BRACS	Aligned	Severe	Diversity	-3.570	[-5.968, -2.334]	<0.0001	0.0004	Tested	-0.059
BRACS	Reversed	Severe	Prevalence	0.948	[0.896, 0.984]	<0.0001	0.0004	Tested	0.794
BRACS	Reversed	Severe	Nominal support	0.880	[0.807, 0.935]	<0.0001	0.0004	Tested	0.736
BRACS	Reversed	Severe	Independent support	-0.388	[-0.855, -0.137]	0.0002	0.0013	Tested	0.111
BRACS	Reversed	Severe	Difficulty	-1.158	[-1.812, -0.685]	<0.0001	0.0004	Tested	-0.059
BRACS	Reversed	Severe	Diversity	-1.060	[-1.951, -0.523]	<0.0001	0.0004	Tested	-0.010
BRACS	Native	Severe (spread)	Prevalence	-0.416	[-2.069, 0.219]	0.2420	0.4841	Tested	0.867
BRACS	Native	Severe (spread)	Nominal support	-0.451	[-2.155, 0.202]	0.5092	0.5092	Tested	0.827
BRACS	Native	Severe (spread)	Independent support	-3.302	[-8.301, -1.464]	<0.0001	0.0004	Tested	0.111
BRACS	Native	Severe (spread)	Difficulty	-4.145	[-9.442, -2.111]	<0.0001	0.0004	Tested	0.021
BRACS	Native	Severe (spread)	Diversity	-3.976	[-10.249, -1.658]	<0.0001	0.0004	Tested	-0.026
BRACS	Aligned	Severe (spread)	Prevalence	0.744	[0.576, 0.841]	<0.0001	0.0004	Tested	0.478
BRACS	Aligned	Severe (spread)	Nominal support	0.745	[0.582, 0.835]	<0.0001	0.0004	Tested	0.467
BRACS	Aligned	Severe (spread)	Independent support	-0.545	[-1.124, -0.170]	<0.0001	0.0004	Tested	-0.139
BRACS	Aligned	Severe (spread)	Difficulty	-0.347	[-0.718, -0.042]	<0.0001	0.0004	Tested	-0.149
BRACS	Aligned	Severe (spread)	Diversity	-0.902	[-1.879, -0.194]	<0.0001	0.0004	Tested	-0.149
BRACS	Reversed	Severe (spread)	Prevalence	0.720	[0.599, 0.806]	<0.0001	0.0004	Tested	0.823
BRACS	Reversed	Severe (spread)	Nominal support	0.664	[0.488, 0.776]	<0.0001	0.0004	Tested	0.790

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Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status	Discrimination R_M
BRACS	Reversed	Severe (spread)	Independent support	-1.112	[-1.955, -0.603]	<0.0001	0.0004	Tested	0.051
BRACS	Reversed	Severe (spread)	Difficulty	-1.615	[-2.556, -0.992]	<0.0001	0.0004	Tested	-0.008
BRACS	Reversed	Severe (spread)	Diversity	-1.013	[-1.895, -0.480]	<0.0001	0.0004	Tested	0.030

Table 10: Recovery of tail-group macro negative log likelihood on the eight calibration-gate-passing units, with paired 95% intervals and Holm-adjusted p -values, reported alongside the discrimination recovery of the same method and unit where that axis also opened.

The split within the family is sharper here than on the discrimination axis. Prevalence-weighted and class-balanced cross-entropy recover between 0.539 and 1.227 of the tail-group likelihood deficit in seven of the eight units, every one at an adjusted $p \leq 0.007$. The remaining three members are negative in all 24 of their cells, and every one of those 24 deteriorations survives Holm adjustment at $p \leq 0.0098$, reaching -4.145 for difficulty weighting under the native severe (spread) condition. A negative recovery fraction here means the method left tail-group probabilities materially worse than deprived cross-entropy had left them, while moving discrimination by an amount indistinguishable from zero. A weight that reads a signal unrelated to the damage does not merely fail to help; it redistributes probability mass across classes and degrades the endpoint the gate identified as damaged.

The native severe (spread) unit is the exception to the prevalence-corner result, and it is worth stating plainly because it is the unit in which spreading the pool changed the character of the damage. There prevalence-weighted and class-balanced cross-entropy return -0.416 and -0.451 with intervals covering zero, so no confirmatory member repaired probability quality on that unit at all, even though both repaired more than four-fifths of its discrimination deficit.

One inference asymmetry should be read from the table rather than around it. The difficulty-aligned moderate calibration contrast carries a bootstrap interval that covers zero, $[-1.4624, 0.0712]$, and a permutation-based adjusted p of 0.0004. The two procedures answer different questions — the interval is a percentile bootstrap over patients, the test a partition-block permutation — and where they disagree the protocol’s decision rule is the test. The disagreement is recorded so that the interval is not read as the test’s confidence statement.

3.4 The wider method roster

The remaining roster methods change the training objective, the sampling distribution, the decision threshold, or the classifier head, and so cannot be compared to the five-family at a fixed intervention mechanism. They are reported with intervals but without confirmatory status: balanced sampling, train-time and post-hoc logit adjustment, classifier retraining, focal loss, class-balanced focal loss, the soft- F_1 and soft-MCC hybrids, and set learning. Their role is to bound what is achievable on these conditions by any tested means, which is what Section 5 needs in order to distinguish a signal that no method exploits from a deficit that no mechanism repairs.

Four of these members must be read as one mechanism rather than four. Balanced sampling, the soft- F_1 hybrid, the soft-MCC hybrid, and the second stage of classifier retraining all resample the training distribution toward uniformity, the last three at a strength that is fixed rather than tuned. Their agreement below is therefore not four independent confirmations, and the tuning record makes the collapse explicit: the search selects the zero-strength point for the soft hybrids in 22 of their 26 selections, at which point they are balanced-sampling cross-entropy with no auxiliary term at all.

Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Native	Balanced	Discrimination	Balanced sampling	-0.208	[-0.894, 0.072]
BRACS	Native	Balanced	Discrimination	CE + soft F_1	-0.034	[-0.489, 0.215]
BRACS	Native	Balanced	Discrimination	CE + soft MCC	-0.497	[-1.996, 0.055]
BRACS	Native	Balanced	Discrimination	CFAL	-0.412	[-1.901, 0.166]
BRACS	Native	Balanced	Discrimination	cRT	-0.734	[-2.240, -0.303]
BRACS	Native	Balanced	Discrimination	Focal loss	0.000	[-0.017, 0.012]
BRACS	Native	Balanced	Discrimination	Logit adjustment (train)	-0.005	[-0.034, 0.018]
BRACS	Native	Balanced	Discrimination	OKO	-0.941	[-3.335, -0.206]
BRACS	Native	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.002	[-0.018, -0.000]
BRACS	Aligned	Balanced	Discrimination	Balanced sampling	-0.209	[-1.011, 0.073]
BRACS	Aligned	Balanced	Discrimination	CE + soft F_1	-0.034	[-0.534, 0.207]
BRACS	Aligned	Balanced	Discrimination	CE + soft MCC	-0.500	[-2.264, 0.053]
BRACS	Aligned	Balanced	Discrimination	CFAL	-0.415	[-2.055, 0.164]
BRACS	Aligned	Balanced	Discrimination	cRT	-0.740	[-2.489, -0.303]
BRACS	Aligned	Balanced	Discrimination	Focal loss	0.000	[-0.018, 0.012]
BRACS	Aligned	Balanced	Discrimination	Logit adjustment (train)	-0.005	[-0.034, 0.019]
BRACS	Aligned	Balanced	Discrimination	OKO	-0.948	[-3.753, -0.197]
BRACS	Aligned	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.002	[-0.017, -0.000]
BRACS	Reversed	Balanced	Discrimination	Balanced sampling	-0.232	[-1.320, 0.089]
BRACS	Reversed	Balanced	Discrimination	CE + soft F_1	-0.038	[-0.643, 0.277]
BRACS	Reversed	Balanced	Discrimination	CE + soft MCC	-0.555	[-2.959, 0.079]
BRACS	Reversed	Balanced	Discrimination	CFAL	-0.460	[-2.630, 0.222]
BRACS	Reversed	Balanced	Discrimination	cRT	-0.821	[-3.518, -0.311]
BRACS	Reversed	Balanced	Discrimination	Focal loss	0.000	[-0.025, 0.014]
BRACS	Reversed	Balanced	Discrimination	Logit adjustment (train)	-0.006	[-0.051, 0.023]
BRACS	Reversed	Balanced	Discrimination	OKO	-1.051	[-5.187, -0.202]
BRACS	Reversed	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.003	[-0.027, -0.000]
BRACS	Native	Moderate	Discrimination	Balanced sampling	0.715	[0.305, 1.033]
BRACS	Native	Moderate	Discrimination	CE + soft F_1	0.714	[0.303, 1.038]
BRACS	Native	Moderate	Discrimination	CE + soft MCC	0.714	[0.303, 1.038]
BRACS	Native	Moderate	Discrimination	CFAL	0.501	[0.150, 0.847]
BRACS	Native	Moderate	Discrimination	cRT	0.381	[-0.132, 0.766]
BRACS	Native	Moderate	Discrimination	Focal loss	0.770	[0.412, 1.078]
BRACS	Native	Moderate	Discrimination	Logit adjustment (train)	0.866	[0.553, 1.237]
BRACS	Native	Moderate	Discrimination	OKO	0.462	[-0.190, 0.988]
BRACS	Native	Moderate	Discrimination	Logit adjustment (post-hoc)	0.234	[0.040, 0.435]
BRACS	Reversed	Moderate	Discrimination	Balanced sampling	0.796	[0.661, 0.930]
BRACS	Reversed	Moderate	Discrimination	CE + soft F_1	0.795	[0.660, 0.930]
BRACS	Reversed	Moderate	Discrimination	CE + soft MCC	0.795	[0.660, 0.930]
BRACS	Reversed	Moderate	Discrimination	CFAL	0.311	[0.112, 0.517]
BRACS	Reversed	Moderate	Discrimination	cRT	0.192	[0.010, 0.354]
BRACS	Reversed	Moderate	Discrimination	Focal loss	0.828	[0.705, 0.957]
BRACS	Reversed	Moderate	Discrimination	Logit adjustment (train)	0.800	[0.658, 0.932]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Reversed	Moderate	Discrimination	OKO	0.742	[0.511, 0.940]
BRACS	Reversed	Moderate	Discrimination	Logit adjustment (post-hoc)	0.233	[0.145, 0.337]
BRACS	Native	Severe	Discrimination	Balanced sampling	0.797	[0.467, 1.067]
BRACS	Native	Severe	Discrimination	CE + soft F_1	0.669	[0.370, 0.901]
BRACS	Native	Severe	Discrimination	CE + soft MCC	0.798	[0.467, 1.069]
BRACS	Native	Severe	Discrimination	CFAL	0.587	[0.357, 0.839]
BRACS	Native	Severe	Discrimination	cRT	0.230	[-0.082, 0.465]
BRACS	Native	Severe	Discrimination	Focal loss	0.725	[0.352, 0.986]
BRACS	Native	Severe	Discrimination	Logit adjustment (train)	0.689	[0.383, 0.948]
BRACS	Native	Severe	Discrimination	OKO	0.518	[0.178, 0.801]
BRACS	Native	Severe	Discrimination	Logit adjustment (post-hoc)	0.385	[0.252, 0.546]
BRACS	Aligned	Severe	Discrimination	Balanced sampling	0.731	[0.408, 1.135]
BRACS	Aligned	Severe	Discrimination	CE + soft F_1	0.674	[0.347, 1.081]
BRACS	Aligned	Severe	Discrimination	CE + soft MCC	0.731	[0.409, 1.135]
BRACS	Aligned	Severe	Discrimination	CFAL	0.383	[0.008, 0.771]
BRACS	Aligned	Severe	Discrimination	cRT	0.052	[-0.351, 0.337]
BRACS	Aligned	Severe	Discrimination	Focal loss	0.598	[0.225, 0.956]
BRACS	Aligned	Severe	Discrimination	Logit adjustment (train)	0.217	[-0.305, 0.554]
BRACS	Aligned	Severe	Discrimination	OKO	0.792	[0.465, 1.274]
BRACS	Aligned	Severe	Discrimination	Logit adjustment (post-hoc)	0.007	[-0.328, 0.216]
BRACS	Reversed	Severe	Discrimination	Balanced sampling	0.777	[0.664, 0.874]
BRACS	Reversed	Severe	Discrimination	CE + soft F_1	0.790	[0.680, 0.894]
BRACS	Reversed	Severe	Discrimination	CE + soft MCC	0.776	[0.662, 0.873]
BRACS	Reversed	Severe	Discrimination	CFAL	0.494	[0.354, 0.628]
BRACS	Reversed	Severe	Discrimination	cRT	0.438	[0.330, 0.565]
BRACS	Reversed	Severe	Discrimination	Focal loss	0.807	[0.699, 0.905]
BRACS	Reversed	Severe	Discrimination	Logit adjustment (train)	0.717	[0.616, 0.827]
BRACS	Reversed	Severe	Discrimination	OKO	0.785	[0.646, 0.929]
BRACS	Reversed	Severe	Discrimination	Logit adjustment (post-hoc)	0.179	[0.124, 0.244]
BRACS	Native	Severe (spread)	Discrimination	Balanced sampling	0.904	[0.763, 1.081]
BRACS	Native	Severe (spread)	Discrimination	CE + soft F_1	0.904	[0.763, 1.081]
BRACS	Native	Severe (spread)	Discrimination	CE + soft MCC	0.904	[0.763, 1.081]
BRACS	Native	Severe (spread)	Discrimination	CFAL	0.485	[0.320, 0.668]
BRACS	Native	Severe (spread)	Discrimination	cRT	0.384	[0.212, 0.544]
BRACS	Native	Severe (spread)	Discrimination	Focal loss	0.859	[0.665, 1.071]
BRACS	Native	Severe (spread)	Discrimination	Logit adjustment (train)	0.720	[0.517, 0.913]
BRACS	Native	Severe (spread)	Discrimination	OKO	0.717	[0.530, 0.916]
BRACS	Native	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	0.612	[0.408, 0.814]
BRACS	Aligned	Severe (spread)	Discrimination	Balanced sampling	0.447	[-0.009, 0.767]
BRACS	Aligned	Severe (spread)	Discrimination	CE + soft F_1	0.447	[-0.009, 0.767]
BRACS	Aligned	Severe (spread)	Discrimination	CE + soft MCC	0.447	[-0.009, 0.767]
BRACS	Aligned	Severe (spread)	Discrimination	CFAL	0.173	[-0.184, 0.515]
BRACS	Aligned	Severe (spread)	Discrimination	cRT	-0.267	[-1.073, 0.068]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Aligned	Severe (spread)	Discrimination	Focal loss	0.387	[-0.192, 0.775]
BRACS	Aligned	Severe (spread)	Discrimination	Logit adjustment (train)	0.128	[-0.598, 0.531]
BRACS	Aligned	Severe (spread)	Discrimination	OKO	0.568	[0.239, 0.967]
BRACS	Aligned	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	-0.403	[-1.478, 0.046]
BRACS	Reversed	Severe (spread)	Discrimination	Balanced sampling	0.853	[0.753, 0.953]
BRACS	Reversed	Severe (spread)	Discrimination	CE + soft F_1	0.853	[0.753, 0.953]
BRACS	Reversed	Severe (spread)	Discrimination	CE + soft MCC	0.853	[0.753, 0.953]
BRACS	Reversed	Severe (spread)	Discrimination	CFAL	0.455	[0.345, 0.577]
BRACS	Reversed	Severe (spread)	Discrimination	cRT	0.427	[0.339, 0.519]
BRACS	Reversed	Severe (spread)	Discrimination	Focal loss	0.829	[0.725, 0.941]
BRACS	Reversed	Severe (spread)	Discrimination	Logit adjustment (train)	0.833	[0.724, 0.953]
BRACS	Reversed	Severe (spread)	Discrimination	OKO	0.832	[0.713, 0.940]
BRACS	Reversed	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	0.462	[0.375, 0.556]
BRACS	Aligned	Moderate	Calibration	Balanced sampling	1.355	[1.096, 2.128]
BRACS	Aligned	Moderate	Calibration	CE + soft F_1	1.355	[1.097, 2.129]
BRACS	Aligned	Moderate	Calibration	CE + soft MCC	1.355	[1.097, 2.129]
BRACS	Aligned	Moderate	Calibration	CFAL	0.529	[-0.212, 0.900]
BRACS	Aligned	Moderate	Calibration	cRT	-1.278	[-4.556, -0.299]
BRACS	Aligned	Moderate	Calibration	Focal loss	1.356	[1.165, 1.900]
BRACS	Aligned	Moderate	Calibration	Logit adjustment (train)	0.745	[0.516, 0.952]
BRACS	Aligned	Moderate	Calibration	OKO	1.165	[0.948, 1.644]
BRACS	Aligned	Moderate	Calibration	Logit adjustment (post-hoc)	-0.874	[-2.066, -0.485]
BRACS	Reversed	Moderate	Calibration	Balanced sampling	0.899	[0.581, 0.981]
BRACS	Reversed	Moderate	Calibration	CE + soft F_1	0.898	[0.582, 0.981]
BRACS	Reversed	Moderate	Calibration	CE + soft MCC	0.898	[0.582, 0.981]
BRACS	Reversed	Moderate	Calibration	CFAL	0.305	[-2.220, 0.735]
BRACS	Reversed	Moderate	Calibration	cRT	-0.461	[-4.132, 0.325]
BRACS	Reversed	Moderate	Calibration	Focal loss	0.898	[0.544, 0.999]
BRACS	Reversed	Moderate	Calibration	Logit adjustment (train)	1.040	[0.944, 1.383]
BRACS	Reversed	Moderate	Calibration	OKO	0.692	[-0.294, 0.878]
BRACS	Reversed	Moderate	Calibration	Logit adjustment (post-hoc)	-0.315	[-1.241, -0.144]
BRACS	Native	Severe	Calibration	Balanced sampling	0.942	[0.805, 1.043]
BRACS	Native	Severe	Calibration	CE + soft F_1	0.838	[0.649, 0.954]
BRACS	Native	Severe	Calibration	CE + soft MCC	0.942	[0.805, 1.043]
BRACS	Native	Severe	Calibration	CFAL	0.474	[-0.201, 0.726]
BRACS	Native	Severe	Calibration	cRT	-0.425	[-1.435, 0.063]
BRACS	Native	Severe	Calibration	Focal loss	0.895	[0.740, 1.001]
BRACS	Native	Severe	Calibration	Logit adjustment (train)	0.980	[0.742, 1.092]
BRACS	Native	Severe	Calibration	OKO	0.813	[0.494, 0.975]
BRACS	Native	Severe	Calibration	Logit adjustment (post-hoc)	-0.541	[-1.094, -0.340]
BRACS	Aligned	Severe	Calibration	Balanced sampling	1.198	[1.093, 1.360]
BRACS	Aligned	Severe	Calibration	CE + soft F_1	1.152	[1.025, 1.283]
BRACS	Aligned	Severe	Calibration	CE + soft MCC	1.198	[1.093, 1.360]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Aligned	Severe	Calibration	CFAL	0.949	[0.763, 1.089]
BRACS	Aligned	Severe	Calibration	cRT	-1.790	[-3.839, -0.771]
BRACS	Aligned	Severe	Calibration	Focal loss	1.213	[1.118, 1.358]
BRACS	Aligned	Severe	Calibration	Logit adjustment (train)	0.620	[0.329, 0.781]
BRACS	Aligned	Severe	Calibration	OKO	1.065	[0.952, 1.190]
BRACS	Aligned	Severe	Calibration	Logit adjustment (post-hoc)	-0.917	[-1.382, -0.637]
BRACS	Reversed	Severe	Calibration	Balanced sampling	0.922	[0.862, 0.959]
BRACS	Reversed	Severe	Calibration	CE + soft F_1	0.931	[0.878, 0.970]
BRACS	Reversed	Severe	Calibration	CE + soft MCC	0.920	[0.861, 0.958]
BRACS	Reversed	Severe	Calibration	CFAL	0.646	[0.448, 0.752]
BRACS	Reversed	Severe	Calibration	cRT	0.177	[-0.218, 0.445]
BRACS	Reversed	Severe	Calibration	Focal loss	0.961	[0.905, 1.003]
BRACS	Reversed	Severe	Calibration	Logit adjustment (train)	0.998	[0.949, 1.041]
BRACS	Reversed	Severe	Calibration	OKO	0.861	[0.766, 0.918]
BRACS	Reversed	Severe	Calibration	Logit adjustment (post-hoc)	-0.357	[-0.484, -0.287]
BRACS	Native	Severe (spread)	Calibration	Balanced sampling	-0.415	[-2.182, 0.225]
BRACS	Native	Severe (spread)	Calibration	CE + soft F_1	-0.415	[-2.182, 0.225]
BRACS	Native	Severe (spread)	Calibration	CE + soft MCC	-0.415	[-2.182, 0.225]
BRACS	Native	Severe (spread)	Calibration	CFAL	-2.417	[-6.493, -0.916]
BRACS	Native	Severe (spread)	Calibration	cRT	-3.759	[-8.219, -2.023]
BRACS	Native	Severe (spread)	Calibration	Focal loss	-0.377	[-2.175, 0.274]
BRACS	Native	Severe (spread)	Calibration	Logit adjustment (train)	0.839	[0.595, 1.169]
BRACS	Native	Severe (spread)	Calibration	OKO	-0.687	[-2.718, 0.077]
BRACS	Native	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Aligned	Severe (spread)	Calibration	Balanced sampling	0.716	[0.522, 0.824]
BRACS	Aligned	Severe (spread)	Calibration	CE + soft F_1	0.716	[0.522, 0.824]
BRACS	Aligned	Severe (spread)	Calibration	CE + soft MCC	0.716	[0.522, 0.824]
BRACS	Aligned	Severe (spread)	Calibration	CFAL	0.262	[-0.235, 0.518]
BRACS	Aligned	Severe (spread)	Calibration	cRT	-0.644	[-1.404, -0.243]
BRACS	Aligned	Severe (spread)	Calibration	Focal loss	0.734	[0.560, 0.833]
BRACS	Aligned	Severe (spread)	Calibration	Logit adjustment (train)	0.919	[0.855, 0.976]
BRACS	Aligned	Severe (spread)	Calibration	OKO	0.728	[0.571, 0.818]
BRACS	Aligned	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Reversed	Severe (spread)	Calibration	Balanced sampling	0.716	[0.598, 0.796]
BRACS	Reversed	Severe (spread)	Calibration	CE + soft F_1	0.716	[0.598, 0.796]
BRACS	Reversed	Severe (spread)	Calibration	CE + soft MCC	0.716	[0.598, 0.796]
BRACS	Reversed	Severe (spread)	Calibration	CFAL	0.343	[0.051, 0.536]
BRACS	Reversed	Severe (spread)	Calibration	cRT	-0.448	[-0.902, -0.190]
BRACS	Reversed	Severe (spread)	Calibration	Focal loss	0.732	[0.614, 0.817]
BRACS	Reversed	Severe (spread)	Calibration	Logit adjustment (train)	0.990	[0.948, 1.030]
BRACS	Reversed	Severe (spread)	Calibration	OKO	0.709	[0.567, 0.805]
BRACS	Reversed	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]

Table 12: Exploratory recovery estimates for the full method roster on gate-passing units, with paired 95% intervals. These methods vary the intervention mechanism as well as the signal, carry no confirmatory status, and are not part of the Holm family.

On the nominal axis the roster confirms that the deficits are repairable. In seven of the eight nominal-axis discrimination units at least one roster method returns more than 0.71 of the deficit, led by balanced sampling, train-time logit adjustment, focal loss, and set learning; the difficulty-aligned severe (spread) unit is the exception, where the best tested mechanism of any kind is set learning at 0.57. On the calibration axis every one of the eight gate-passing units has a method reaching at least 0.72, with train-time logit adjustment the most consistent, and no method overshoots its reference by a margin the intervals cannot absorb.

On the independent-support axis the roster fails as completely as the confirmatory family. In all three balanced units not one of the nine exploratory methods returns a positive recovery whose interval excludes zero, and several are reliably harmful: classifier retraining returns -0.73 to -0.82 and set learning -0.94 to -1.05 , both with intervals lying entirely below zero. Taken together with Section 3.1, this identifies a materially damaged regime in which no tested mechanism, signal-matched or otherwise, repairs the damage. Four of the eleven discrimination-gate units — the three independent-support units and difficulty-aligned severe (spread) — have no tested method reaching two-thirds of their deficit.

4 Explaining Heterogeneity

Damage magnitude and recovery headroom vary across conditions, and RQ3 asks which realized shortages track that variation. Two association models are specified, one for signed balanced-accuracy damage over the cross-entropy cells and one for recovery over the gate-passing method cells. Both use the same linear predictor over standardized quantities,

$$\mu_j = \alpha + u_{g[j]} + \beta_\rho \log \rho_j + \beta_{\text{ind}} S_{\text{ind},j} + \beta_A A_j + \beta_{\text{div}} S_{\text{div},j}, \quad (1)$$

where ρ_j is achieved severity, $S_{\text{ind},j}$ the independent-support shortage, A_j the support–difficulty alignment, $S_{\text{div},j}$ the diversity shortage, and $u_{g[j]}$ a random intercept over dataset–target groups that absorbs level differences between datasets. Observation errors come from the bootstrap distributions of the corresponding endpoints.

The BRACS cells are complete, but the combined models await the planned TCGA-UT benchmark because the between-dataset variance is not identifiable from one dataset–target group. The recorded BRACS cells allow the combined fit to be assembled without repeating its bootstrap. Binary targets remain excluded because support–difficulty alignment over two classes is always ± 1 and would confound the pooled coefficient.

Placeholder: TCGA-UT estimates and the combined RQ3 fit will be added here after that benchmark is complete.

What can be said without the models is what Section 2.2 establishes descriptively. Damage rises monotonically with achieved severity under all three assignments; it is roughly three times larger under the difficulty-reversed assignment than under the aligned one at matched severity; and the independent-support axis produces damage at a fifth to a quarter of the nominal dose, which is the single observation in this report that a nominal-only account of shortage cannot absorb.

5 Discussion

The preceding sections establish what the controlled allocations damaged and how much of that damage the tested methods returned. This section reads those outcomes as design evidence. The question it answers is not which existing method won, but what a mitigation method would have to read, and to do, in order to address the shortages this dataset actually presents.

5.1 What the evidence supports

On RQ1, both manipulated axes damage the endpoints, and they damage them differently. Nominal deprivation opened the discrimination gate in eight of its nine units and the probability-quality gate in eight, with damage growing in severity and in assignment adversity, and landing on the body tier rather than the tail wherever the assignment gives the easiest classes the least support. Independent-support deprivation, at a fifth to a quarter of the nominal dose and with patch counts held exactly equal, opened the discrimination gate in all three of its units at deficits of 2.6–3.0 percentage points, and fell just under the probability-quality threshold at 0.132–0.135 nats. The two axes are therefore not redundant: a class can be measurably underserved with no shortfall in its patch count at all.

On RQ2, recovery separates the confirmatory family into two groups along the same split in every unit where anything recovers, and the split is by signal, not by intervention strength. Prevalence-weighted and class-balanced cross-entropy return most of both deficits on the nominal axis; independent-support, difficulty, and diversity weighting return essentially none of the discrimination deficit and degrade tail-group probability quality in all 24 of their calibration cells, every one surviving multiplicity adjustment. The matched- β diagnostic excludes tuned weight strength as the explanation. On the independent-support axis nothing recovers: not the matched member, not the prevalence corner, and not one of the nine exploratory mechanisms. The direct matched-versus-unmatched contrast is 0.06 percentage points of balanced accuracy with an interval covering zero.

RQ3 remains pending until the TCGA-UT results and combined fit are added in Section 4.

5.2 Which shortage a mitigation method must read

A method must estimate the shortage that binds, so the empirical question is whether the dominant shortage was identifiable and whether reading it helped. On this dataset the dominant shortage was identifiable in nine of 12 units, and reading it helped only where it was nominal support. The more informative case is the one in which the dominant shortage is independent support and reading it still does not help.

That case is a real measurement rather than a structural artefact. In a design where a single per-class evidence pool serves every condition, independent support cannot vary, and a method built to read it is being tested on conditions in which the quantity it reads is constant — a null under those circumstances says nothing. The spread arm removes that objection: independent support varies by 0.41 to 0.67 nats between the two arms, the pools are nested so the manipulation adds patients rather than exchanging them, and the resulting deficit is large enough to open a prespecified gate. Independent-support weighted cross-entropy then recovers 0.013 to 0.015 of it. The inference the evidence supports is that a mean-one class weight computed from patient counts does not repair damage caused by a shortage of patients, at this dose, on this dataset.

The mechanism that failure suggests is worth stating because it constrains method design directly. All five confirmatory members apply a per-class scalar to the loss, which can only change how heavily each class’s examples are emphasized. When a class lacks patients rather than patches, its patches are correlated: they carry fewer independent observations of the class than their count implies, and no reweighting of that count adds information the sample does not contain. A method that addresses independent-support shortage must therefore do something other than reweight — draw on evidence outside the class’s own examples, or change what is estimated rather than how heavily it is weighted.

The frequency of the ambiguous designation qualifies the routing question separately. Three of 12 units, all at severe severity, have their two largest shortage scores within a quarter of a standard deviation. A taxonomy that cannot separate the top two axes in a quarter of its conditions does not partition sharply enough to route a method by selecting among signals,

which points toward a method that combines them.

5.3 Signal against mechanism

Because the five confirmatory methods share one intervention mechanism and differ only in the signal their weights read, the spread of recovery across them isolates how much leverage the choice of signal carries. That spread is wide on the nominal axis, running from below zero to 0.892 within single units, so signal choice determines whether a weighted-loss method works at all there. The spread is not, however, evidence that a better shortage estimate would help, because it is bimodal rather than graded: two members work almost everywhere on that axis and three work almost nowhere, with little in between.

The independent-support axis separates signal from mechanism more cleanly than the nominal axis can, because there the mechanism is held fixed across the five members and the wider roster fails alongside them. Nine exploratory methods spanning resampling, logit adjustment, decision-threshold correction, classifier retraining, and set learning were applied to a deficit of roughly three balanced-accuracy points, and none returned a positive fraction of it whose interval excludes zero. When both the signal contrast and the mechanism contrast come back null on the same units, the limitation lies not in the choice among tested designs but in what all of them share.

The roster’s harmful members sharpen that reading. Classifier retraining and set learning are the two mechanisms that most aggressively rebalance the effective training distribution, and they are also the two that damage the independent-support units most, at -0.73 to -1.05 . Rebalancing toward classes whose examples come from few patients amplifies those patients’ idiosyncrasies rather than the class’s signal, which is the failure mode a patient-aware method would have to avoid by construction.

5.4 Where recovery ceilings

The units that define an unmet need are those in which a gate opened and no method, confirmatory or exploratory, returned a substantial fraction of the deficit. On this dataset there are four of them, out of eleven discrimination-gate units. Three are the independent-support units, where nothing recovers at all. The fourth is difficulty-aligned severe (spread), where the best tested mechanism of any kind reaches 0.57. A fifth case sits on the other axis: the native severe (spread) unit, where both prevalence-corner members return negative probability-quality recovery even though both repair most of the same unit’s discrimination deficit.

This is the finding that most changes what a new method would be for. The ceiling on the nominal axis is set by neither the data nor the fixed representation — existing mechanisms reach most of that damage, and a new method cannot be motivated there by damage left unrepaired. The ceiling on the independent-support axis is set by the mechanism the whole roster shares. A method motivated by this benchmark is therefore a method for the second regime, and its claim would be a claim about patient-level evidence rather than about class counts.

The complementary case remains part of the specification. The difficulty-aligned moderate condition damaged probability quality without a certifiable discrimination deficit, and the three independent-support units damaged discrimination without a certifiable probability-quality deficit. A method that intervenes on the wrong axis spends capacity and risks degrading an endpoint that was not damaged, and Section 3.3 shows that risk is real: in all 24 cells where a non-prevalence signal was applied to a calibration-damaged unit, tail-group probability quality fell below the untreated reference. Knowing when, and on which axis, not to act is part of the specification rather than an afterthought.

5.5 Design requirements and open questions

The preceding subsections resolve into a set of requirements for a mitigation method, each traceable to a specific result, annotated with the strength of evidence behind it and, where relevant, with what a second dataset would have to confirm.

1. Where the shortage is nominal, the method must read prevalence or nominal support, and must not rely on a difficulty or diversity estimate as its primary signal. *Confirmatory*, from the five-family separation in Table 5 across the eight nominal-axis discrimination units.
2. It must not degrade tail-group probability quality while repairing discrimination. *Confirmatory*, from the 24 negative calibration-recovery cells in Table 10, all of which survive Holm adjustment.
3. It must decline to intervene where allocation produced no material deficit on the axis it targets, which on this dataset is the difficulty-aligned moderate condition on discrimination and the three independent-support units on probability quality. *Confirmatory*, from Sections 2.2 and 2.3.
4. Independent-support shortage requires a mechanism other than per-class loss reweighting. *Confirmatory within this dataset*, from the joint null of the confirmatory family and the exploratory roster on the three balanced units. *Open across datasets*: this cohort carries the weakest independent dose the candidate datasets offer, so a replication at a stronger dose is needed before the null is read as a property of reweighting rather than of this dose.
5. It must not require access to test prevalence, labelled data beyond the training partition, or per-condition tuning that a deployment could not perform. *Design constraint*, carried from the protocol rather than established by a result here.
6. Whether the shortage estimate must be formed per condition rather than transferred across datasets. *Pending TCGA-UT*: this is the requirement RQ3’s variance components were specified to settle.

Requirements four and six will be reassessed after the TCGA-UT analysis.

5.6 Pending TCGA-UT analysis

Placeholder: this subsection will report whether the independent-support result replicates on TCGA-UT, compare dataset-specific materiality thresholds, and interpret the combined RQ3 model. Binary and low-cardinality targets remain outside that model because support–difficulty alignment over two classes is structurally fixed at ± 1 .

6 Conclusion and Scope

The current BRACS results answer the first two protocol questions. Controlled allocation damages tail-class discrimination and probability quality along two axes that ordinary imbalance confounds: nominal patch deprivation and independent patient deprivation. Signal-matched mitigation recovers most nominal damage through prevalence and nominal-support weighting, but no tested signal or mechanism recovers the independent-support damage. Section 5 converts this evidence into requirements for what a mitigation method must read, what it must not degrade, and where it must decline to act. RQ3 will be completed with TCGA-UT.

Four conditions bound the current claims. The three partitions are repeated partitions of BRACS, not independent cohorts, so the intervals quantify variation across partitioning and

initialization rather than across populations. The three independent-support units share one deprived allocation and differ only in their reference assignment, so they are correlated readings rather than three replications. The benchmark covers patch classification only, where each prediction unit carries its own label; at whole-slide level, reallocating labelled slides changes nominal and independent support together. Temperature scaling is reported as a diagnostic and never determines whether recovery is analysed.

A Realized Conditions and Analysis Validity

This appendix carries the execution record that Section 2 summarizes in prose. It documents the realized construction and whether the conditions required for confirmatory inference were met.

A.1 Realized construction

Table 14 reports the achieved imbalance ratio and normalized-entropy deficit for each partition and controlled condition, together with the realized patient, slide, and patch counts. Conditions falling outside the tolerance bands, $\rho \in [9, 11]$ for moderate and $\rho \in [90, 110]$ for severe, are flagged as off-target, and allocations that collapsed to balanced are flagged as degenerate and excluded from all estimands. No condition carries either flag. Patient and slide counts are unique within a realization: one patient may supply patches to more than one class, so per-class counts are not summed.

Dataset	Split	Condition	Assignment	Units	Slides	Patches	ρ	$1 - H_{\text{norm}}$	Tolerance
BRACS	0	Natural	—	103	271	217399	5.901	0.1202	—
BRACS	0	Balanced	—	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced	Aligned	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced (spread)	Aligned	103	209	16796	1.000	0.0000	—
BRACS	0	Moderate	Aligned	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Aligned	85	194	16796	100.467	0.3530	On target
BRACS	0	Severe (spread)	Aligned	103	200	16796	100.467	0.3530	On target
BRACS	0	Balanced	Reversed	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced (spread)	Reversed	103	209	16796	1.000	0.0000	—
BRACS	0	Moderate	Reversed	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Reversed	85	203	16796	100.467	0.3530	On target
BRACS	0	Severe (spread)	Reversed	103	209	16796	100.467	0.3530	On target
BRACS	0	Balanced	Native	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced (spread)	Native	103	209	16796	1.000	0.0000	—
BRACS	0	Moderate	Native	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Native	85	204	16796	100.467	0.3530	On target
BRACS	0	Severe (spread)	Native	103	209	16796	100.467	0.3530	On target
BRACS	1	Natural	—	103	273	193010	5.722	0.1178	—
BRACS	1	Balanced	—	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced	Aligned	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced (spread)	Aligned	103	229	27202	1.000	0.0000	—
BRACS	1	Moderate	Aligned	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Aligned	89	232	27202	100.301	0.3530	On target
BRACS	1	Severe (spread)	Aligned	103	229	27202	100.301	0.3530	On target
BRACS	1	Balanced	Reversed	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced (spread)	Reversed	103	229	27202	1.000	0.0000	—
BRACS	1	Moderate	Reversed	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Reversed	89	233	27202	100.301	0.3530	On target
BRACS	1	Severe (spread)	Reversed	103	228	27202	100.301	0.3530	On target
BRACS	1	Balanced	Native	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced (spread)	Native	103	229	27202	1.000	0.0000	—
BRACS	1	Moderate	Native	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Native	89	233	27202	100.301	0.3530	On target
BRACS	1	Severe (spread)	Native	103	229	27202	100.301	0.3530	On target
BRACS	2	Natural	—	103	262	217704	8.297	0.1432	—
BRACS	2	Balanced	—	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced	Aligned	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced (spread)	Aligned	103	193	13982	1.001	0.0000	—
BRACS	2	Moderate	Aligned	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Aligned	78	173	13982	100.360	0.3530	On target
BRACS	2	Severe (spread)	Aligned	103	187	13982	100.360	0.3530	On target
BRACS	2	Balanced	Reversed	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced (spread)	Reversed	103	193	13982	1.001	0.0000	—

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Dataset	Split	Condition	Assignment	Units	Slides	Patches	ρ	$1 - H_{\text{norm}}$	Tolerance
BRACS	2	Moderate	Reversed	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Reversed	78	179	13982	100.360	0.3530	On target
BRACS	2	Severe (spread)	Reversed	103	193	13982	100.360	0.3530	On target
BRACS	2	Balanced	Native	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced (spread)	Native	103	193	13982	1.001	0.0000	—
BRACS	2	Moderate	Native	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Native	78	179	13982	100.360	0.3530	On target
BRACS	2	Severe (spread)	Native	103	193	13982	100.360	0.3530	On target

Table 14: Realized training support per partition and controlled condition: unique patients, slides, patches, achieved ρ , and normalized-entropy deficit, with tolerance status. The natural anchor and the two balanced references carry no severity tolerance.

The two rows that define the independent-support axis are the balanced and balanced (spread) entries within each partition. They carry identical patch counts and an identical ρ of 1.000, and differ only in patient count, 85 against 103, and correspondingly in slides. That the spread arm reaches the same 103 patients the natural condition draws on confirms that it exhausts the eligible inventory, so the dose reported in Section 2.1 is the largest this cohort can supply under the contribution caps.

A.2 Resampling validity and completeness

The label-only preflight resamples patients before any model is fitted and checks whether class-level endpoints can be estimated stably at all. Three concentration checks apply per class: whether the 2.5th percentile of the Kish effective count falls below five, whether a single patient carries more than half the weight in more than 5% of replicates, and whether a pooled cell draws on fewer than five contributing patients. A class failing any check designates the dataset descriptive-only, which forces both gates closed.

Dataset	Class	Kish n_{eff}	2.5th pct.	Max unit weight	Dominant-unit share	Descriptive-only
BRACS	ADH	10.8	7.2	0.180	0.000	No
BRACS	DCIS	8.8	5.5	0.211	0.000	No
BRACS	FEA	9.3	6.0	0.201	0.000	No
BRACS	IC	14.3	9.7	0.145	0.000	No
BRACS	N	14.4	9.9	0.144	0.000	No
BRACS	PB	18.9	13.5	0.116	0.000	No
BRACS	UDH	17.9	12.6	0.121	0.000	No

Table 16: Label-only bootstrap preflight per class: Kish effective count and its 2.5th percentile, maximum single-patient weight fraction, share of replicates with a dominant patient, and the resulting descriptive-only designation. Values are the most adverse across the three partitions.

No class is designated descriptive-only on the pooled evidence cell that every gated quantity uses. The tightest margin is ductal carcinoma in situ, whose 2.5th-percentile Kish effective count is 5.5 against a floor of five, and no class records a dominant patient in any replicate. The margin is narrow because the cohort is small: the seven classes draw on 85 to 103 patients in total, which is the same scarcity the independent-support axis manipulates.

That scarcity binds one level down. Evaluated per partition and per class rather than pooled, the preflight flags the ADH, DCIS, and FEA cells as descriptive-only on all three partitions, with 2.5th-percentile Kish effective counts between 1.8 and 2.7 and dominant-patient shares up to 0.32. No gated quantity in this report is computed on those cells, since every gate and every recovery estimate reads the pooled cell, but the per-class, per-partition entries of Appendix C rest on them and are descriptive for that reason.

Completeness is enforced rather than reported after the fact. Every confirmation block must contain all five initialization seeds, every expected combination of tail assignment, condition, method, and gate must appear in all three partitions, and the achieved severity of a condition must not vary by more than half across partitions.

Dataset	Splits	Seeds per block	Comparisons	Max severity spread	Descriptive-only	Guards
BRACS	3	5	378	0.002	No	All passed

Table 17: Completeness and consistency guards: confirmation-seed blocks, expected comparison coverage across the three partitions, and achieved-severity spread. These guards halt aggregation rather than degrade it, so a written result implies every guard passed.

A.3 Budget, tuning selections, and deviations

Training budget is expressed in example presentations rather than optimizer updates, because methods consume different amounts of data per update and an update budget would confound the method comparison with data exposure. The frozen budget is $E = 503,880$ example presentations for every controlled condition and $E = 6,521,970$ for the natural anchor; each method derives its own update count as $\lceil E/e_m \rceil$, where e_m is its examples per update. The check-point interval is rescaled to that derived count so that every method receives approximately 170 validation passes regardless of how many updates its budget yields.

Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Balanced	—	Balanced sampling	0.0001	0.7500	No	503880
BRACS	Balanced	—	CE	3e-05	—	No	503880
BRACS	Balanced	—	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced	—	CE + soft MCC	0.0003	16.0000	No	503880
BRACS	Balanced	—	CFAL	0.0001	0.2500	No	503880
BRACS	Balanced	—	Nominal support	3e-05	0.9999	No	503880
BRACS	Balanced	—	cRT	0.01	—	No	503880
BRACS	Balanced	—	Focal loss	3e-05	0.0000	No	503880
BRACS	Balanced	—	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced	—	Logit adjustment (train)	3e-05	2.0000	No	503880
BRACS	Balanced	—	OKO	0.003	2.0000	No	503880
BRACS	Balanced	—	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced	—	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Balanced	—	Diversity	3e-05	0.2500	No	503880
BRACS	Balanced	—	Prevalence	3e-05	0.7500	No	503880
BRACS	Balanced (spread)	Aligned	Balanced sampling	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	CE	3e-05	—	No	503880
BRACS	Balanced (spread)	Aligned	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Aligned	CE + soft MCC	0.0001	16.0000	No	503880
BRACS	Balanced (spread)	Aligned	CFAL	0.001	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	Nominal support	3e-05	0.9990	No	503880
BRACS	Balanced (spread)	Aligned	cRT	0.003	—	No	503880
BRACS	Balanced (spread)	Aligned	Focal loss	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Aligned	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced (spread)	Aligned	Logit adjustment (train)	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	OKO	0.003	1.0000	No	503880
BRACS	Balanced (spread)	Aligned	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	Logit adjustment (post-hoc)	—	2.0000	No	503880
BRACS	Balanced (spread)	Aligned	Diversity	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	Prevalence	3e-05	0.7500	No	503880
BRACS	Balanced (spread)	Reversed	Balanced sampling	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	CE	3e-05	—	No	503880
BRACS	Balanced (spread)	Reversed	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Reversed	CE + soft MCC	0.0001	16.0000	No	503880
BRACS	Balanced (spread)	Reversed	CFAL	0.001	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	Nominal support	3e-05	0.9990	No	503880
BRACS	Balanced (spread)	Reversed	cRT	0.003	—	No	503880
BRACS	Balanced (spread)	Reversed	Focal loss	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Reversed	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced (spread)	Reversed	Logit adjustment (train)	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	OKO	0.003	1.0000	No	503880
BRACS	Balanced (spread)	Reversed	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	Logit adjustment (post-hoc)	—	2.0000	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Balanced (spread)	Reversed	Diversity	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	Prevalence	3e-05	0.7500	No	503880
BRACS	Balanced (spread)	Native	Balanced sampling	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	CE	3e-05	—	No	503880
BRACS	Balanced (spread)	Native	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Native	CE + soft MCC	0.0001	16.0000	No	503880
BRACS	Balanced (spread)	Native	CFAL	0.001	0.2500	No	503880
BRACS	Balanced (spread)	Native	Nominal support	3e-05	0.9990	No	503880
BRACS	Balanced (spread)	Native	cRT	0.003	—	No	503880
BRACS	Balanced (spread)	Native	Focal loss	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Native	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced (spread)	Native	Logit adjustment (train)	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	OKO	0.003	1.0000	No	503880
BRACS	Balanced (spread)	Native	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	Logit adjustment (post-hoc)	—	2.0000	No	503880
BRACS	Balanced (spread)	Native	Diversity	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	Prevalence	3e-05	0.7500	No	503880
BRACS	Moderate	Aligned	Balanced sampling	0.0001	1.0000	No	503880
BRACS	Moderate	Aligned	CE	0.003	—	No	503880
BRACS	Moderate	Aligned	CE + soft F_1	0.0001	0.0000	No	503880
BRACS	Moderate	Aligned	CE + soft MCC	0.0001	0.0000	No	503880
BRACS	Moderate	Aligned	CFAL	0.003	0.2500	No	503880
BRACS	Moderate	Aligned	Nominal support	0.0001	0.9999	No	503880
BRACS	Moderate	Aligned	cRT	0.01	—	No	503880
BRACS	Moderate	Aligned	Focal loss	0.0001	0.5000	No	503880
BRACS	Moderate	Aligned	Independent support	0.001	0.9900	No	503880
BRACS	Moderate	Aligned	Logit adjustment (train)	0.0001	1.0000	No	503880
BRACS	Moderate	Aligned	OKO	0.001	4.0000	No	503880
BRACS	Moderate	Aligned	Difficulty	0.001	0.2500	No	503880
BRACS	Moderate	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Moderate	Aligned	Diversity	0.003	1.0000	No	503880
BRACS	Moderate	Aligned	Prevalence	0.0001	1.0000	No	503880
BRACS	Moderate	Reversed	Balanced sampling	0.0001	1.0000	No	503880
BRACS	Moderate	Reversed	CE	0.003	—	No	503880
BRACS	Moderate	Reversed	CE + soft F_1	0.0001	0.0000	No	503880
BRACS	Moderate	Reversed	CE + soft MCC	0.0001	0.0000	No	503880
BRACS	Moderate	Reversed	CFAL	0.003	0.2500	No	503880
BRACS	Moderate	Reversed	Nominal support	0.0001	0.9999	No	503880
BRACS	Moderate	Reversed	cRT	0.01	—	No	503880
BRACS	Moderate	Reversed	Focal loss	0.0001	0.5000	No	503880
BRACS	Moderate	Reversed	Independent support	0.001	0.9900	No	503880
BRACS	Moderate	Reversed	Logit adjustment (train)	0.0001	1.0000	No	503880
BRACS	Moderate	Reversed	OKO	0.001	4.0000	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Moderate	Reversed	Difficulty	0.001	0.2500	No	503880
BRACS	Moderate	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Moderate	Reversed	Diversity	0.003	1.0000	No	503880
BRACS	Moderate	Reversed	Prevalence	0.0001	1.0000	No	503880
BRACS	Moderate	Native	Balanced sampling	0.0001	1.0000	No	503880
BRACS	Moderate	Native	CE	0.003	—	No	503880
BRACS	Moderate	Native	CE + soft F_1	0.0001	0.0000	No	503880
BRACS	Moderate	Native	CE + soft MCC	0.0001	0.0000	No	503880
BRACS	Moderate	Native	CFAL	0.003	0.2500	No	503880
BRACS	Moderate	Native	Nominal support	0.0001	0.9999	No	503880
BRACS	Moderate	Native	cRT	0.01	—	No	503880
BRACS	Moderate	Native	Focal loss	0.0001	0.5000	No	503880
BRACS	Moderate	Native	Independent support	0.001	0.9900	No	503880
BRACS	Moderate	Native	Logit adjustment (train)	0.0001	1.0000	No	503880
BRACS	Moderate	Native	OKO	0.001	4.0000	No	503880
BRACS	Moderate	Native	Difficulty	0.001	0.2500	No	503880
BRACS	Moderate	Native	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Moderate	Native	Diversity	0.003	1.0000	No	503880
BRACS	Moderate	Native	Prevalence	0.0001	1.0000	No	503880
BRACS	Natural	—	CE	3e-05	—	No	6521970
BRACS	Severe	Aligned	Balanced sampling	1e-05	1.0000	No	503880
BRACS	Severe	Aligned	CE	0.001	—	No	503880
BRACS	Severe	Aligned	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe	Aligned	CE + soft MCC	1e-05	0.0000	No	503880
BRACS	Severe	Aligned	CFAL	0.003	1.0000	No	503880
BRACS	Severe	Aligned	Nominal support	0.0003	0.9999	No	503880
BRACS	Severe	Aligned	cRT	0.01	—	No	503880
BRACS	Severe	Aligned	Focal loss	3e-05	0.0000	No	503880
BRACS	Severe	Aligned	Independent support	0.001	0.9990	No	503880
BRACS	Severe	Aligned	Logit adjustment (train)	0.0003	1.0000	No	503880
BRACS	Severe	Aligned	OKO	0.001	4.0000	No	503880
BRACS	Severe	Aligned	Difficulty	0.001	0.5000	No	503880
BRACS	Severe	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Severe	Aligned	Diversity	0.003	0.5000	No	503880
BRACS	Severe	Aligned	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe	Reversed	Balanced sampling	1e-05	1.0000	No	503880
BRACS	Severe	Reversed	CE	0.001	—	No	503880
BRACS	Severe	Reversed	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe	Reversed	CE + soft MCC	1e-05	0.0000	No	503880
BRACS	Severe	Reversed	CFAL	0.003	1.0000	No	503880
BRACS	Severe	Reversed	Nominal support	0.0003	0.9999	No	503880
BRACS	Severe	Reversed	cRT	0.01	—	No	503880
BRACS	Severe	Reversed	Focal loss	3e-05	0.0000	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Severe	Reversed	Independent support	0.001	0.9990	No	503880
BRACS	Severe	Reversed	Logit adjustment (train)	0.0003	1.0000	No	503880
BRACS	Severe	Reversed	OKO	0.001	4.0000	No	503880
BRACS	Severe	Reversed	Difficulty	0.001	0.5000	No	503880
BRACS	Severe	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Severe	Reversed	Diversity	0.003	0.5000	No	503880
BRACS	Severe	Reversed	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe	Native	Balanced sampling	1e-05	1.0000	No	503880
BRACS	Severe	Native	CE	0.001	—	No	503880
BRACS	Severe	Native	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe	Native	CE + soft MCC	1e-05	0.0000	No	503880
BRACS	Severe	Native	CFAL	0.003	1.0000	No	503880
BRACS	Severe	Native	Nominal support	0.0003	0.9999	No	503880
BRACS	Severe	Native	cRT	0.01	—	No	503880
BRACS	Severe	Native	Focal loss	3e-05	0.0000	No	503880
BRACS	Severe	Native	Independent support	0.001	0.9990	No	503880
BRACS	Severe	Native	Logit adjustment (train)	0.0003	1.0000	No	503880
BRACS	Severe	Native	OKO	0.001	4.0000	No	503880
BRACS	Severe	Native	Difficulty	0.001	0.5000	No	503880
BRACS	Severe	Native	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Severe	Native	Diversity	0.003	0.5000	No	503880
BRACS	Severe	Native	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Balanced sampling	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Aligned	CE	0.001	—	No	503880
BRACS	Severe (spread)	Aligned	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Aligned	CE + soft MCC	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Aligned	CFAL	0.001	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Nominal support	0.0001	0.9999	No	503880
BRACS	Severe (spread)	Aligned	cRT	0.01	—	No	503880
BRACS	Severe (spread)	Aligned	Focal loss	3e-05	0.5000	No	503880
BRACS	Severe (spread)	Aligned	Independent support	0.001	0.9900	No	503880
BRACS	Severe (spread)	Aligned	Logit adjustment (train)	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Aligned	OKO	0.0003	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Difficulty	0.001	0.2500	No	503880
BRACS	Severe (spread)	Aligned	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Diversity	0.001	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Balanced sampling	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Reversed	CE	0.001	—	No	503880
BRACS	Severe (spread)	Reversed	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Reversed	CE + soft MCC	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Reversed	CFAL	0.001	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Nominal support	0.0001	0.9999	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Severe (spread)	Reversed	cRT	0.01	—	No	503880
BRACS	Severe (spread)	Reversed	Focal loss	3e-05	0.5000	No	503880
BRACS	Severe (spread)	Reversed	Independent support	0.001	0.9900	No	503880
BRACS	Severe (spread)	Reversed	Logit adjustment (train)	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Reversed	OKO	0.0003	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Difficulty	0.001	0.2500	No	503880
BRACS	Severe (spread)	Reversed	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Diversity	0.001	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Native	Balanced sampling	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Native	CE	0.001	—	No	503880
BRACS	Severe (spread)	Native	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Native	CE + soft MCC	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Native	CFAL	0.001	1.0000	No	503880
BRACS	Severe (spread)	Native	Nominal support	0.0001	0.9999	No	503880
BRACS	Severe (spread)	Native	cRT	0.01	—	No	503880
BRACS	Severe (spread)	Native	Focal loss	3e-05	0.5000	No	503880
BRACS	Severe (spread)	Native	Independent support	0.001	0.9900	No	503880
BRACS	Severe (spread)	Native	Logit adjustment (train)	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Native	OKO	0.0003	1.0000	No	503880
BRACS	Severe (spread)	Native	Difficulty	0.001	0.2500	No	503880
BRACS	Severe (spread)	Native	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Severe (spread)	Native	Diversity	0.001	1.0000	No	503880
BRACS	Severe (spread)	Native	Prevalence	3e-05	1.0000	No	503880

Table 19: Selected learning rate and strength control per condition, tail assignment, and method, with the frozen example-presentation budget and whether the selection terminated at a learning-rate envelope boundary.

None of the 196 selections terminated at a learning-rate envelope boundary, so in no case did the envelope rather than the validation data determine the choice. Three properties of the selection procedure are stated here as limits on the claims that rest on it rather than as defects.

The selection objective is natural-prevalence validation balanced accuracy, tie-broken by macro F_1 and then by negative log likelihood. Tail-group negative log likelihood is a co-primary endpoint with its own gate, and no method was ever tuned for it. Every statement in Section 3.3 about a method degrading probability quality is therefore a statement about a method optimized for a different endpoint.

Selection also pools across tail assignments: one configuration is chosen per condition and method and applied to all three assignments, which the table records as repeated rows, and all 60 condition–method cells that span more than one assignment carry identical selections. Since the assignment determines which classes need help, this pooling systematically dilutes the difficulty- and diversity-reading methods relative to a per-assignment selection, and their null results should be read with that in mind.

The tuning budget itself differs by method: 16 candidates for most, four for cross-entropy and classifier retraining, and a single shard for post-hoc logit adjustment, which requires no training of its own.

Three further properties of the selections change what a method label denotes and are recorded for that reason. The search selects the zero-strength point in 29 of the 196 selections: 13 for the soft- F_1 hybrid, nine for the soft-MCC hybrid, and seven for focal loss. At zero strength the soft hybrids reduce to balanced-sampling cross-entropy, because their auxiliary term vanishes while their forced balanced sampler does not, and focal loss reduces to class-weighted cross-entropy, because its class-aware α term applies at unit strength independently of γ . Where those selections were made, the roster row reports the reduced method rather than the named one.

One consequence of that reduction is a defect in the search rather than in the fitted models, and it is recorded rather than repaired because repairing it would change the tuning protocol between this dataset and any replication. The search treats focal loss as anchored to plain cross-entropy, so its $\gamma = 0$ candidate is scored using cross-entropy’s already-computed validation metrics instead of being trained. Because α applies unconditionally, that candidate is in fact class-weighted cross-entropy and not plain cross-entropy, so the seven selections that landed on $\gamma = 0$ were made against the wrong metrics. The confirmation runs train the correct objective, and focal loss is an exploratory roster member that enters no confirmatory test, so no gated or Holm-adjusted quantity in this report depends on it; its recovery estimates in Table 12 should nonetheless be read as estimates of a configuration that was not selected on its own evidence.

Two deviations from the locked protocol were made before any definitive gate row was read. The two deficit-gate thresholds were derived rather than assumed: the protocol had fixed discrimination at 2 percentage points and calibration at 0.05 nats without a stated derivation, while the discrimination value conflicted with the support pilot’s 1-percentage-point materiality constant for the same endpoint under the same kind of manipulation. Both were replaced by quantities the protocol already fixes: $\max(1, 2\sigma_{\text{seed}}) = 1.443$ percentage points for discrimination and $2\sigma_{\text{seed}} = 0.14782$ nats for calibration, where σ_{seed} is the confirmation-seed standard deviation of the corresponding balanced-condition endpoint on this dataset. Separately, the gate-defining estimator is case-macro rather than patch-micro, and the evidence cell is pooled across a case’s partition appearances rather than taken per partition; both correct a mismatch between a diagnostic and the estimator it certifies, since the bootstrap draws one shared weight per case over the union of the three test frames.

A.4 Method diagnostics

Semantic-scale weighted cross-entropy falls back to a raw weight of 1, that is, to no reweighting, for any class whose matched draw is degenerate at reweighting time. The `ssb_invalid_draws`

counter records how often that fallback fires; a nonzero count would be grounds to distrust the diversity-weighted recovery readings for the affected condition.

Dataset	Split	Condition	Method	Seeds	Ssb invalid draws	Ssb pools
BRACS	0	Balanced (spread)	Diversity	15	0	15
BRACS	0	Moderate	Diversity	15	0	15
BRACS	0	Severe	Diversity	15	0	15
BRACS	0	Severe (spread)	Diversity	15	0	15
BRACS	0	Balanced	Diversity	5	0	5
BRACS	1	Balanced (spread)	Diversity	15	0	15
BRACS	1	Moderate	Diversity	15	0	15
BRACS	1	Severe	Diversity	15	0	15
BRACS	1	Severe (spread)	Diversity	15	0	15
BRACS	1	Balanced	Diversity	5	0	5
BRACS	2	Balanced (spread)	Diversity	15	0	15
BRACS	2	Moderate	Diversity	15	0	15
BRACS	2	Severe	Diversity	15	0	15
BRACS	2	Severe (spread)	Diversity	15	0	15
BRACS	2	Balanced	Diversity	5	0	5

Table 21: Per-partition, per-condition `method_diagnostics` counters for semantic-scale weighted cross-entropy: degenerate-draw fallback count against the number of reweighting pools formed, summed across confirmation seeds.

The fallback never fired: 0 invalid draws against 195 reweighting pools across the three partitions. The diversity-weighted results in Section 3 therefore measure the intended weighting rather than a degenerate fallback.

B Aggregate Damage Estimates

Tables 22 and 23 provide the assignment-level estimates summarized in Sections 2.2 and 2.3.

Dataset	Assignment	Severity	D_{BA} (pp)	95% CI (pp)	Gate
BRACS	Native	Balanced	2.95	[1.04, 5.25]	Open
BRACS	Aligned	Balanced	2.93	[0.91, 5.22]	Open
BRACS	Reversed	Balanced	2.64	[0.55, 5.04]	Open
BRACS	Native	Moderate	3.34	[1.72, 5.09]	Open
BRACS	Aligned	Moderate	1.50	[-0.40, 3.39]	Interval covers zero
BRACS	Reversed	Moderate	8.17	[6.09, 10.60]	Open
BRACS	Native	Severe	5.45	[3.27, 7.81]	Open
BRACS	Aligned	Severe	4.30	[2.08, 6.59]	Open
BRACS	Reversed	Severe	13.23	[10.78, 15.84]	Open
BRACS	Native	Severe (spread)	8.47	[5.82, 11.42]	Open
BRACS	Aligned	Severe (spread)	4.04	[1.85, 6.53]	Open
BRACS	Reversed	Severe (spread)	13.79	[11.37, 16.34]	Open

Table 22: Balanced-accuracy deficit with paired 95% intervals.

Dataset	Assignment	Severity	D_{cal} (nats)	95% CI	Gate	ECE	ECE 95% CI
BRACS	Native	Balanced	0.133	[0.060, 0.211]	Below threshold	0.051	[0.040, 0.087]
BRACS	Aligned	Balanced	0.135	[0.065, 0.211]	Below threshold	0.051	[0.040, 0.087]
BRACS	Reversed	Balanced	0.132	[0.059, 0.207]	Below threshold	0.051	[0.040, 0.087]
BRACS	Native	Moderate	0.186	[-0.007, 0.397]	Interval covers zero	0.148	[0.103, 0.210]
BRACS	Aligned	Moderate	0.379	[0.157, 0.715]	Open	0.143	[0.108, 0.189]
BRACS	Reversed	Moderate	0.754	[0.180, 1.706]	Open	0.215	[0.170, 0.276]
BRACS	Native	Severe	0.688	[0.332, 1.099]	Open	0.220	[0.176, 0.276]
BRACS	Aligned	Severe	0.729	[0.488, 1.045]	Open	0.219	[0.174, 0.280]
BRACS	Reversed	Severe	1.744	[1.208, 2.295]	Open	0.313	[0.256, 0.384]
BRACS	Native	Severe (spread)	0.201	[0.095, 0.331]	Open	0.147	[0.112, 0.198]
BRACS	Aligned	Severe (spread)	0.903	[0.574, 1.258]	Open	0.209	[0.160, 0.275]
BRACS	Reversed	Severe (spread)	1.127	[0.827, 1.473]	Open	0.259	[0.208, 0.327]

Table 23: Tail-group negative-log-likelihood deficit with paired 95% intervals and secondary expected calibration error.

C Per-Class and Tier Endpoints

Section 2.4 reports damage at tier level; this appendix gives the per-class detail behind it. Recall, F_1 , negative log likelihood, and Brier score are listed per class and condition for the cross-entropy reference, averaged within a partition and then with equal weight across the three partitions, with the tier assignment given alongside. The table is restricted to cross-entropy because that is the reference against which every deficit in Section 2 is defined. As recorded in Appendix A.2, the per-partition cells for the ADH, DCIS, and FEA classes are designated descriptive-only by the preflight, so their entries here carry no inferential weight.

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Aligned	Balanced (spread)	ADH	body	0.046	0.052	2.487	1.091
BRACS	Aligned	Balanced (spread)	ADH	tail	0.211	0.216	2.825	1.217
BRACS	Aligned	Balanced (spread)	DCIS	body	0.480	0.504	0.804	0.365
BRACS	Aligned	Balanced (spread)	DCIS	head	0.476	0.368	1.038	0.541
BRACS	Aligned	Balanced (spread)	DCIS	tail	0.633	0.653	1.199	0.628
BRACS	Aligned	Balanced (spread)	FEA	body	0.439	0.307	3.524	1.346
BRACS	Aligned	Balanced (spread)	FEA	head	0.442	0.374	1.900	0.850
BRACS	Aligned	Balanced (spread)	IC	head	0.880	0.904	0.397	0.168
BRACS	Aligned	Balanced (spread)	N	head	0.669	0.472	1.069	0.574
BRACS	Aligned	Balanced (spread)	PB	tail	0.453	0.575	0.881	0.483
BRACS	Aligned	Balanced (spread)	UDH	tail	0.419	0.340	1.909	0.929
BRACS	Aligned	Moderate	ADH	body	0.186	0.121	3.311	1.200
BRACS	Aligned	Moderate	ADH	tail	0.037	0.053	3.249	1.313
BRACS	Aligned	Moderate	DCIS	body	0.557	0.490	0.907	0.385
BRACS	Aligned	Moderate	DCIS	head	0.614	0.411	1.242	0.490
BRACS	Aligned	Moderate	DCIS	tail	0.403	0.523	1.137	0.519
BRACS	Aligned	Moderate	FEA	body	0.385	0.195	4.278	1.429
BRACS	Aligned	Moderate	FEA	head	0.545	0.268	2.371	0.898
BRACS	Aligned	Moderate	IC	head	0.883	0.892	0.646	0.234
BRACS	Aligned	Moderate	N	head	0.840	0.431	1.111	0.576
BRACS	Aligned	Moderate	PB	tail	0.186	0.293	1.496	0.732
BRACS	Aligned	Moderate	UDH	tail	0.206	0.242	2.063	0.951
BRACS	Aligned	Severe	ADH	body	0.246	0.136	5.255	1.306
BRACS	Aligned	Severe	ADH	tail	0.020	0.034	3.521	1.316
BRACS	Aligned	Severe	DCIS	body	0.514	0.472	1.534	0.571
BRACS	Aligned	Severe	DCIS	head	0.568	0.410	1.192	0.504
BRACS	Aligned	Severe	DCIS	tail	0.241	0.356	1.743	0.702
BRACS	Aligned	Severe	FEA	body	0.469	0.180	3.554	1.289
BRACS	Aligned	Severe	FEA	head	0.497	0.235	3.545	1.030
BRACS	Aligned	Severe	IC	head	0.889	0.882	0.919	0.266
BRACS	Aligned	Severe	N	head	0.832	0.406	1.485	0.684
BRACS	Aligned	Severe	PB	tail	0.133	0.222	1.913	0.844
BRACS	Aligned	Severe	UDH	tail	0.129	0.154	2.407	0.995
BRACS	Aligned	Severe (spread)	ADH	body	0.308	0.196	3.679	1.228
BRACS	Aligned	Severe (spread)	ADH	tail	0.021	0.038	5.225	1.440
BRACS	Aligned	Severe (spread)	DCIS	body	0.511	0.464	1.448	0.529
BRACS	Aligned	Severe (spread)	DCIS	head	0.390	0.258	2.465	0.806
BRACS	Aligned	Severe (spread)	DCIS	tail	0.197	0.310	1.493	0.640
BRACS	Aligned	Severe (spread)	FEA	body	0.406	0.195	5.132	1.317
BRACS	Aligned	Severe (spread)	FEA	head	0.478	0.249	3.601	1.036
BRACS	Aligned	Severe (spread)	IC	head	0.904	0.886	0.824	0.234
BRACS	Aligned	Severe (spread)	N	head	0.862	0.435	1.273	0.571
BRACS	Aligned	Severe (spread)	PB	tail	0.184	0.287	1.688	0.759

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Aligned	Severe (spread)	UDH	tail	0.133	0.167	2.884	1.101
BRACS	Reversed	Balanced (spread)	ADH	body	0.061	0.063	2.401	1.071
BRACS	Reversed	Balanced (spread)	ADH	head	0.230	0.220	2.768	1.195
BRACS	Reversed	Balanced (spread)	DCIS	body	0.449	0.485	0.819	0.375
BRACS	Reversed	Balanced (spread)	DCIS	head	0.616	0.642	1.216	0.636
BRACS	Reversed	Balanced (spread)	DCIS	tail	0.504	0.382	0.990	0.512
BRACS	Reversed	Balanced (spread)	FEA	body	0.419	0.299	3.561	1.333
BRACS	Reversed	Balanced (spread)	FEA	tail	0.436	0.371	1.904	0.852
BRACS	Reversed	Balanced (spread)	IC	tail	0.876	0.903	0.419	0.174
BRACS	Reversed	Balanced (spread)	N	tail	0.673	0.466	1.075	0.575
BRACS	Reversed	Balanced (spread)	PB	head	0.447	0.571	0.882	0.483
BRACS	Reversed	Balanced (spread)	UDH	head	0.415	0.338	1.900	0.923
BRACS	Reversed	Moderate	ADH	body	0.063	0.063	2.708	1.080
BRACS	Reversed	Moderate	ADH	head	0.441	0.180	3.601	1.308
BRACS	Reversed	Moderate	DCIS	body	0.329	0.401	1.816	0.606
BRACS	Reversed	Moderate	DCIS	head	0.832	0.457	1.255	0.628
BRACS	Reversed	Moderate	DCIS	tail	0.364	0.347	1.871	0.686
BRACS	Reversed	Moderate	FEA	body	0.198	0.169	6.328	1.495
BRACS	Reversed	Moderate	FEA	tail	0.305	0.278	3.072	0.904
BRACS	Reversed	Moderate	IC	tail	0.743	0.825	0.835	0.229
BRACS	Reversed	Moderate	N	tail	0.311	0.329	1.583	0.683
BRACS	Reversed	Moderate	PB	head	0.348	0.449	2.411	0.902
BRACS	Reversed	Moderate	UDH	head	0.464	0.296	2.643	0.995
BRACS	Reversed	Severe	ADH	body	0.094	0.090	5.539	1.247
BRACS	Reversed	Severe	ADH	head	0.553	0.173	4.036	1.281
BRACS	Reversed	Severe	DCIS	body	0.240	0.300	2.604	0.740
BRACS	Reversed	Severe	DCIS	head	0.826	0.337	1.675	0.564
BRACS	Reversed	Severe	DCIS	tail	0.313	0.334	2.878	0.856
BRACS	Reversed	Severe	FEA	body	0.126	0.103	8.673	1.575
BRACS	Reversed	Severe	FEA	tail	0.201	0.214	4.261	1.076
BRACS	Reversed	Severe	IC	tail	0.605	0.732	1.182	0.348
BRACS	Reversed	Severe	N	tail	0.222	0.270	2.518	0.865
BRACS	Reversed	Severe	PB	head	0.364	0.463	3.632	1.022
BRACS	Reversed	Severe	UDH	head	0.447	0.273	3.589	1.095
BRACS	Reversed	Severe (spread)	ADH	body	0.051	0.053	4.132	1.247
BRACS	Reversed	Severe (spread)	ADH	head	0.583	0.178	3.659	1.191
BRACS	Reversed	Severe (spread)	DCIS	body	0.145	0.195	1.704	0.645
BRACS	Reversed	Severe (spread)	DCIS	head	0.823	0.342	1.540	0.591
BRACS	Reversed	Severe (spread)	DCIS	tail	0.195	0.192	3.120	0.952
BRACS	Reversed	Severe (spread)	FEA	body	0.204	0.164	6.270	1.461
BRACS	Reversed	Severe (spread)	FEA	tail	0.192	0.204	3.394	1.048
BRACS	Reversed	Severe (spread)	IC	tail	0.562	0.706	0.984	0.345
BRACS	Reversed	Severe (spread)	N	tail	0.202	0.266	1.836	0.742

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Reversed	Severe (spread)	PB	head	0.465	0.538	2.277	0.855
BRACS	Reversed	Severe (spread)	UDH	head	0.409	0.263	3.562	1.143
BRACS	Native	Balanced (spread)	ADH	tail	0.174	0.173	2.662	1.157
BRACS	Native	Balanced (spread)	DCIS	tail	0.533	0.507	1.005	0.507
BRACS	Native	Balanced (spread)	FEA	body	0.447	0.349	2.422	1.005
BRACS	Native	Balanced (spread)	IC	tail	0.873	0.902	0.426	0.176
BRACS	Native	Balanced (spread)	N	head	0.680	0.467	1.043	0.555
BRACS	Native	Balanced (spread)	PB	head	0.451	0.573	0.893	0.492
BRACS	Native	Balanced (spread)	UDH	head	0.403	0.341	1.933	0.936
BRACS	Native	Moderate	ADH	tail	0.082	0.089	2.972	1.246
BRACS	Native	Moderate	DCIS	tail	0.443	0.461	1.098	0.550
BRACS	Native	Moderate	FEA	body	0.443	0.310	2.739	1.031
BRACS	Native	Moderate	IC	tail	0.826	0.871	0.524	0.199
BRACS	Native	Moderate	N	head	0.731	0.428	1.248	0.644
BRACS	Native	Moderate	PB	head	0.319	0.430	1.706	0.768
BRACS	Native	Moderate	UDH	head	0.426	0.326	2.054	0.914
BRACS	Native	Severe	ADH	tail	0.095	0.104	3.842	1.315
BRACS	Native	Severe	DCIS	tail	0.395	0.442	1.424	0.591
BRACS	Native	Severe	FEA	body	0.417	0.316	3.337	1.104
BRACS	Native	Severe	IC	tail	0.742	0.824	0.663	0.234
BRACS	Native	Severe	N	head	0.764	0.409	1.341	0.589
BRACS	Native	Severe	PB	head	0.293	0.390	2.779	0.967
BRACS	Native	Severe	UDH	head	0.421	0.306	2.470	0.928
BRACS	Native	Severe (spread)	ADH	tail	0.118	0.119	3.014	1.201
BRACS	Native	Severe (spread)	DCIS	tail	0.259	0.309	1.281	0.636
BRACS	Native	Severe (spread)	FEA	body	0.376	0.331	3.003	1.110
BRACS	Native	Severe (spread)	IC	tail	0.719	0.817	0.402	0.184
BRACS	Native	Severe (spread)	N	head	0.748	0.421	1.369	0.617
BRACS	Native	Severe (spread)	PB	head	0.401	0.469	1.425	0.719
BRACS	Native	Severe (spread)	UDH	head	0.403	0.298	2.369	0.990

Table 25: Equal-weight three-partition per-class test endpoints for the cross-entropy reference by condition, with the tier assignment each condition induces.

D Calibration Detail

The calibration gate reads tail-group macro negative log likelihood, but the shape of the miscalibration matters for method design and is not visible in a scalar. This appendix reports the raw and temperature-scaled expected calibration error, negative log likelihood, and Brier score per condition and method for the cross-entropy reference and the confirmatory family, and shows the reliability curves from which they are computed.

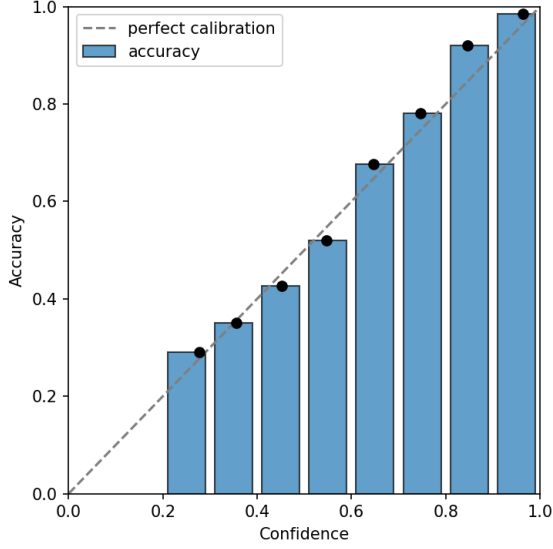
Dataset	Assignment	Condition	Method	NLL	Brier	ECE		Temp. NLL	Temp. Brier	Temp. ECE
BRACS	Aligned	Balanced (spread)	CE	0.918	0.453	0.040	[0.031, 0.078]	0.920	0.454	0.046
BRACS	Aligned	Balanced (spread)	Nominal support	0.950	0.471	0.041	[0.032, 0.073]	0.945	0.470	0.038
BRACS	Aligned	Balanced (spread)	Independent support	0.952	0.472	0.040	[0.032, 0.072]	0.948	0.471	0.037
BRACS	Aligned	Balanced (spread)	Difficulty	0.955	0.471	0.042	[0.033, 0.073]	0.949	0.470	0.037
BRACS	Aligned	Balanced (spread)	Diversity	0.955	0.473	0.042	[0.032, 0.074]	0.950	0.472	0.038
BRACS	Aligned	Balanced (spread)	Prevalence	0.951	0.471	0.041	[0.032, 0.073]	0.946	0.470	0.038
BRACS	Aligned	Moderate	CE	1.211	0.548	0.143	[0.108, 0.189]	1.047	0.510	0.059
BRACS	Aligned	Moderate	Nominal support	1.099	0.534	0.083	[0.060, 0.130]	1.066	0.524	0.062
BRACS	Aligned	Moderate	Independent support	1.223	0.578	0.170	[0.122, 0.230]	1.087	0.529	0.072
BRACS	Aligned	Moderate	Difficulty	1.426	0.601	0.185	[0.130, 0.257]	1.119	0.545	0.088
BRACS	Aligned	Moderate	Diversity	1.385	0.614	0.205	[0.151, 0.276]	1.125	0.546	0.082
BRACS	Aligned	Moderate	Prevalence	1.130	0.543	0.091	[0.067, 0.139]	1.080	0.529	0.064
BRACS	Aligned	Severe	CE	1.551	0.621	0.219	[0.174, 0.280]	1.098	0.531	0.065
BRACS	Aligned	Severe	Nominal support	1.210	0.572	0.150	[0.105, 0.211]	1.098	0.535	0.075
BRACS	Aligned	Severe	Independent support	2.165	0.744	0.320	[0.249, 0.408]	1.191	0.567	0.099
BRACS	Aligned	Severe	Difficulty	1.930	0.736	0.312	[0.243, 0.401]	1.199	0.575	0.110
BRACS	Aligned	Severe	Diversity	2.617	0.784	0.349	[0.279, 0.438]	1.225	0.577	0.098
BRACS	Aligned	Severe	Prevalence	1.089	0.531	0.081	[0.060, 0.130]	1.063	0.526	0.073
BRACS	Aligned	Severe (spread)	CE	1.564	0.593	0.209	[0.160, 0.275]	1.064	0.514	0.064
BRACS	Aligned	Severe (spread)	Nominal support	1.000	0.498	0.072	[0.046, 0.116]	0.996	0.497	0.072
BRACS	Aligned	Severe (spread)	Independent support	2.304	0.749	0.328	[0.258, 0.418]	1.179	0.570	0.110
BRACS	Aligned	Severe (spread)	Difficulty	2.114	0.724	0.310	[0.241, 0.400]	1.162	0.565	0.105
BRACS	Aligned	Severe (spread)	Diversity	2.454	0.742	0.327	[0.253, 0.420]	1.165	0.561	0.101
BRACS	Aligned	Severe (spread)	Prevalence	0.971	0.484	0.055	[0.039, 0.094]	0.973	0.486	0.060
BRACS	Reversed	Balanced (spread)	CE	0.919	0.453	0.041	[0.030, 0.077]	0.923	0.454	0.046
BRACS	Reversed	Balanced (spread)	Nominal support	0.960	0.475	0.044	[0.033, 0.077]	0.953	0.474	0.039
BRACS	Reversed	Balanced (spread)	Independent support	0.964	0.476	0.048	[0.036, 0.081]	0.954	0.474	0.040
BRACS	Reversed	Balanced (spread)	Difficulty	0.982	0.483	0.055	[0.041, 0.087]	0.970	0.480	0.044
BRACS	Reversed	Balanced (spread)	Diversity	0.964	0.476	0.048	[0.036, 0.079]	0.955	0.475	0.040
BRACS	Reversed	Balanced (spread)	Prevalence	0.964	0.476	0.047	[0.035, 0.080]	0.956	0.475	0.040
BRACS	Reversed	Moderate	CE	1.769	0.635	0.215	[0.170, 0.276]	1.164	0.548	0.062
BRACS	Reversed	Moderate	Nominal support	1.087	0.527	0.068	[0.056, 0.104]	1.065	0.518	0.045
BRACS	Reversed	Moderate	Independent support	1.585	0.642	0.188	[0.154, 0.235]	1.224	0.579	0.081
BRACS	Reversed	Moderate	Difficulty	2.287	0.749	0.268	[0.227, 0.319]	1.322	0.624	0.088
BRACS	Reversed	Moderate	Diversity	2.301	0.743	0.270	[0.220, 0.327]	1.306	0.615	0.074
BRACS	Reversed	Moderate	Prevalence	1.095	0.529	0.067	[0.047, 0.109]	1.067	0.520	0.049
BRACS	Reversed	Severe	CE	2.582	0.751	0.313	[0.256, 0.384]	1.271	0.590	0.067
BRACS	Reversed	Severe	Nominal support	1.130	0.545	0.086	[0.065, 0.126]	1.104	0.534	0.062
BRACS	Reversed	Severe	Independent support	2.513	0.801	0.307	[0.261, 0.362]	1.364	0.651	0.093
BRACS	Reversed	Severe	Difficulty	3.392	0.920	0.401	[0.345, 0.463]	1.453	0.693	0.085
BRACS	Reversed	Severe	Diversity	3.102	0.857	0.352	[0.295, 0.415]	1.408	0.666	0.079
BRACS	Reversed	Severe	Prevalence	1.064	0.518	0.064	[0.050, 0.101]	1.052	0.514	0.055
BRACS	Reversed	Severe (spread)	CE	1.915	0.680	0.259	[0.208, 0.327]	1.180	0.563	0.048

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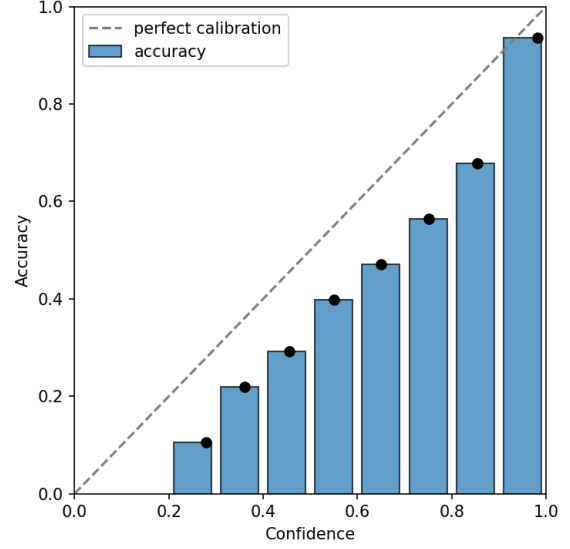
Dataset	Assignment	Condition	Method	NLL	Brier	ECE		Temp. NLL	Temp. Brier	Temp. ECE
BRACS	Reversed	Severe (spread)	Nominal support	1.037	0.509	0.074	[0.049, 0.108]	1.028	0.505	0.067
BRACS	Reversed	Severe (spread)	Independent support	2.653	0.864	0.358	[0.310, 0.414]	1.391	0.676	0.037
BRACS	Reversed	Severe (spread)	Difficulty	3.073	0.889	0.381	[0.332, 0.437]	1.405	0.682	0.049
BRACS	Reversed	Severe (spread)	Diversity	2.635	0.856	0.354	[0.306, 0.410]	1.391	0.674	0.037
BRACS	Reversed	Severe (spread)	Prevalence	1.007	0.494	0.061	[0.043, 0.091]	1.003	0.495	0.066
BRACS	Native	Balanced (spread)	CE	0.925	0.457	0.050	[0.035, 0.084]	0.925	0.456	0.049
BRACS	Native	Balanced (spread)	Nominal support	0.960	0.474	0.055	[0.040, 0.084]	0.951	0.472	0.044
BRACS	Native	Balanced (spread)	Independent support	0.960	0.474	0.055	[0.040, 0.085]	0.950	0.471	0.042
BRACS	Native	Balanced (spread)	Difficulty	0.967	0.477	0.053	[0.038, 0.082]	0.959	0.475	0.042
BRACS	Native	Balanced (spread)	Diversity	0.965	0.477	0.055	[0.042, 0.086]	0.954	0.474	0.043
BRACS	Native	Balanced (spread)	Prevalence	0.963	0.476	0.054	[0.040, 0.084]	0.954	0.474	0.045
BRACS	Native	Moderate	CE	1.274	0.564	0.148	[0.103, 0.210]	1.111	0.527	0.077
BRACS	Native	Moderate	Nominal support	1.023	0.504	0.061	[0.050, 0.092]	1.018	0.502	0.051
BRACS	Native	Moderate	Independent support	1.185	0.549	0.117	[0.080, 0.185]	1.091	0.525	0.067
BRACS	Native	Moderate	Difficulty	1.267	0.584	0.128	[0.085, 0.194]	1.150	0.555	0.069
BRACS	Native	Moderate	Diversity	1.355	0.592	0.137	[0.091, 0.205]	1.153	0.555	0.058
BRACS	Native	Moderate	Prevalence	1.031	0.506	0.058	[0.048, 0.095]	1.027	0.505	0.053
BRACS	Native	Severe	CE	1.793	0.638	0.220	[0.176, 0.276]	1.199	0.554	0.079
BRACS	Native	Severe	Nominal support	1.131	0.545	0.100	[0.067, 0.150]	1.089	0.530	0.062
BRACS	Native	Severe	Independent support	1.751	0.665	0.208	[0.143, 0.297]	1.259	0.599	0.084
BRACS	Native	Severe	Difficulty	2.330	0.734	0.271	[0.215, 0.343]	1.310	0.620	0.090
BRACS	Native	Severe	Diversity	2.085	0.714	0.247	[0.192, 0.320]	1.311	0.616	0.077
BRACS	Native	Severe	Prevalence	1.072	0.522	0.078	[0.053, 0.122]	1.051	0.517	0.060
BRACS	Native	Severe (spread)	CE	1.203	0.549	0.147	[0.112, 0.198]	1.062	0.514	0.062
BRACS	Native	Severe (spread)	Nominal support	0.979	0.487	0.054	[0.044, 0.086]	0.971	0.485	0.045
BRACS	Native	Severe (spread)	Independent support	1.369	0.629	0.161	[0.119, 0.219]	1.192	0.583	0.044
BRACS	Native	Severe (spread)	Difficulty	1.483	0.650	0.194	[0.148, 0.250]	1.199	0.589	0.041
BRACS	Native	Severe (spread)	Diversity	1.472	0.647	0.176	[0.128, 0.235]	1.208	0.592	0.041
BRACS	Native	Severe (spread)	Prevalence	1.000	0.491	0.067	[0.054, 0.100]	0.980	0.487	0.053
BRACS	—	Balanced	CE	0.993	0.486	0.051	[0.040, 0.087]	0.995	0.487	0.055
BRACS	—	Balanced	Nominal support	1.057	0.516	0.057	[0.043, 0.101]	1.040	0.511	0.045
BRACS	—	Balanced	Independent support	1.055	0.516	0.057	[0.043, 0.102]	1.038	0.511	0.045
BRACS	—	Balanced	Difficulty	1.057	0.516	0.058	[0.043, 0.105]	1.039	0.510	0.044
BRACS	—	Balanced	Diversity	1.045	0.511	0.048	[0.039, 0.087]	1.035	0.512	0.048
BRACS	—	Balanced	Prevalence	1.057	0.516	0.057	[0.042, 0.101]	1.040	0.511	0.045
BRACS	—	Natural	CE	0.925	0.443	0.069	[0.043, 0.107]	0.896	0.435	0.042

Table 27: Raw and temperature-scaled expected calibration error, negative log likelihood, and Brier score per condition and method, with crossed patient-bootstrap intervals on the calibration error. Temperature is fitted on natural-prevalence validation data and never gates.

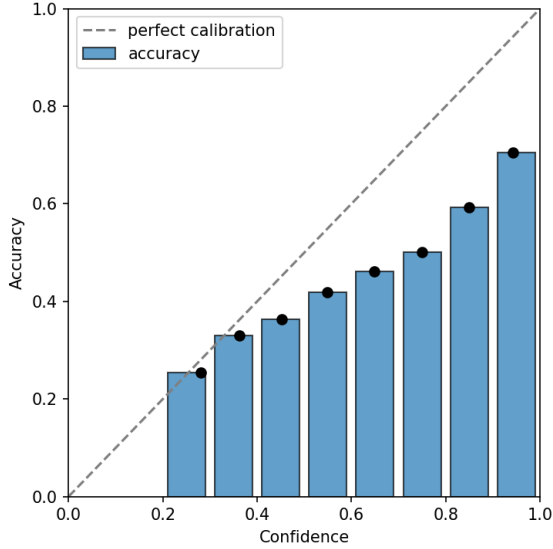
Temperature scaling removes most of the observed miscalibration on the cross-entropy reference wherever the calibration gate opened: under the difficulty-reversed severe condition it takes expected calibration error from 0.313 to 0.067 and negative log likelihood from 2.582 to 1.271. It does not substitute for mitigation, because a single global scalar cannot redistribute probability mass between classes and therefore leaves the tail-group deficit that the gate reads largely intact. This is why the protocol treats it as a diagnostic and never lets it gate. The two balanced references are the informative control: their calibration error is 0.040 to 0.051 before scaling, and scaling moves it by at most 0.006, so the miscalibration the controlled conditions introduce is a product of the allocation rather than a standing property of the representation.



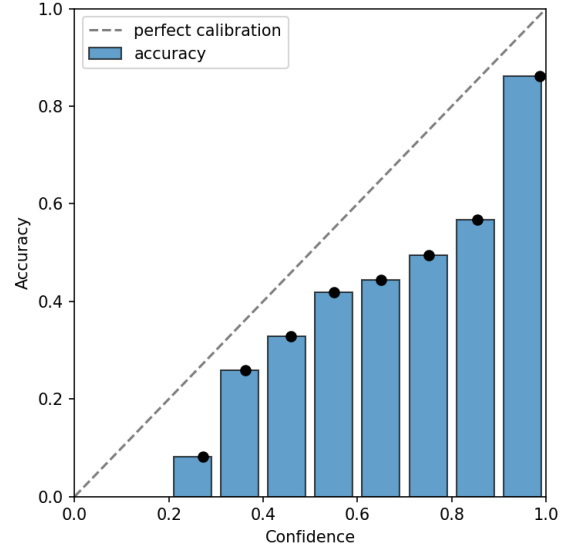
(a) Balanced reference, all classes



(b) Severe, native assignment, tail classes



(c) Severe, aligned assignment, tail classes



(d) Severe, reversed assignment, tail classes

Figure 1: Reliability curves for the balanced reference and for the tail classes under the severe condition of each tail assignment, with bin accuracy against mean predicted confidence over ten equal-width bins, on the first locked partition.

E Per-Partition Results

The main text averages the three partitions with equal weight, which is only defensible if no single partition drives a conclusion. This appendix reports the cross-entropy deficits separately for each partition, alongside the spread of achieved severity across them.

Dataset	Assignment	Severity	Gate	Split 0	Split 1	Split 2	Equal weight	ρ spread
BRACS	Native	Balanced	Discrimination	0.0252	0.0064	0.0570	0.0295	0.001
BRACS	Native	Balanced	Calibration	0.0702	0.1625	0.1671	0.1333	0.001
BRACS	Aligned	Balanced	Discrimination	0.0252	0.0030	0.0597	0.0293	0.001
BRACS	Aligned	Balanced	Calibration	0.0699	0.1350	0.2015	0.1355	0.001
BRACS	Reversed	Balanced	Discrimination	0.0200	0.0053	0.0540	0.0264	0.001
BRACS	Reversed	Balanced	Calibration	0.0597	0.1322	0.2044	0.1321	0.001
BRACS	Native	Moderate	Discrimination	0.0367	0.0353	0.0283	0.0334	0.000
BRACS	Native	Moderate	Calibration	0.0274	0.0921	0.4374	0.1856	0.000
BRACS	Aligned	Moderate	Discrimination	0.0163	0.0297	-0.0012	0.0150	0.000
BRACS	Aligned	Moderate	Calibration	0.5975	0.2429	0.2978	0.3794	0.000
BRACS	Reversed	Moderate	Discrimination	0.0861	0.0676	0.0914	0.0817	0.000
BRACS	Reversed	Moderate	Calibration	0.4098	0.0372	1.8136	0.7535	0.000
BRACS	Native	Severe	Discrimination	0.0467	0.0599	0.0569	0.0545	0.002
BRACS	Native	Severe	Calibration	0.4832	0.1602	1.4204	0.6879	0.002
BRACS	Aligned	Severe	Discrimination	0.0119	0.0730	0.0441	0.0430	0.002
BRACS	Aligned	Severe	Calibration	0.3062	0.9779	0.9042	0.7294	0.002
BRACS	Reversed	Severe	Discrimination	0.1288	0.1287	0.1394	0.1323	0.002
BRACS	Reversed	Severe	Calibration	1.2663	1.5668	2.3976	1.7436	0.002
BRACS	Native	Severe (spread)	Discrimination	0.0563	0.0987	0.0991	0.0847	0.002
BRACS	Native	Severe (spread)	Calibration	0.3566	0.1748	0.0704	0.2006	0.002
BRACS	Aligned	Severe (spread)	Discrimination	0.0556	0.0348	0.0309	0.0404	0.002
BRACS	Aligned	Severe (spread)	Calibration	1.1440	0.6894	0.8769	0.9034	0.002
BRACS	Reversed	Severe (spread)	Discrimination	0.1429	0.1166	0.1541	0.1379	0.002
BRACS	Reversed	Severe (spread)	Calibration	1.4890	0.9357	0.9561	1.1269	0.002

Table 29: Cross-entropy deficits by partition on both gate axes, with the equal-weight average and the relative achieved-severity spread across partitions, per condition and tail assignment. No gate decision rests on a single partition, but the independent-support axis is the least uniform across them and this qualifies its reading. Its three units clear the 1.44-percentage-point discrimination threshold in two of the three partitions and fall to 0.30–0.64 percentage points on the second, against 5.40–5.97 on the third; the equal-weight average that opens the gate is therefore carried by two partitions rather than three. On the nominal axis the pattern is steadier: every gate-passing unit clears the threshold in all three partitions except difficulty-aligned severe, whose smallest partition-level deficit is 1.19 percentage points. On the calibration axis the three balanced units are consistent in falling below threshold, and among the eight gate-passing units the reversed moderate condition is the most variable, ranging from 0.037–1.814 nats. Achieved severity itself varies across partitions by at most 0.002 in relative terms, well inside the 0.5 consistency guard.

The conclusions of Section 3 are less exposed to this variation than the gate decisions are, because the recovery estimates on the independent-support axis are null in every partition rather than null only on average. The qualification attaches to the size of the independent-support deficit, not to the finding that nothing recovers it.

F Computational Cost

Cost bears on the design question because a method whose recovery requires substantially more computation than the reference is a different proposition from one that matches it. This appendix records processed examples, unique exposures, derived updates, examples per update, accelerator time, peak memory, and model size for every roster method, expressed as a matched difference against cross-entropy in the same condition.

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Balanced (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Aligned	Balanced (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Aligned	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Balanced (spread)	cRT	Peak memory (bytes)	1.319e+07	[1.319e+07, 1.319e+07]
BRACS	Aligned	Balanced (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Balanced (spread)	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Balanced (spread)	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Aligned	Balanced (spread)	OKO	Peak memory (bytes)	-6.161e+05	[-6.161e+05, -6.161e+05]
BRACS	Aligned	Balanced (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Aligned	Balanced (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Aligned	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Aligned	Balanced (spread)	Diversity	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Diversity	Peak memory (bytes)	5.247e+07	[5.225e+07, 5.274e+07]
BRACS	Aligned	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Aligned	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.002]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Moderate	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	CFAL	Accelerator hours	0.004	[0.004, 0.005]
BRACS	Aligned	Moderate	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Aligned	Moderate	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Moderate	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Moderate	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
BRACS	Aligned	Moderate	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Moderate	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Moderate	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Focal loss	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Aligned	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Aligned	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	OKO	Accelerator hours	0.001	[0, 0.001]
BRACS	Aligned	Moderate	OKO	Peak memory (bytes)	-3.54e+05	[-3.54e+05, -3.54e+05]
BRACS	Aligned	Moderate	OKO	Processed examples	2983	[2983, 2983]
BRACS	Aligned	Moderate	OKO	Parameters	3591	[3591, 3591]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Moderate	OKO	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Aligned	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Moderate	Diversity	Peak memory (bytes)	-1.147e+07	[-1.152e+07, -1.141e+07]
BRACS	Aligned	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe	CE + soft F_1	Accelerator hours	0.005	[0.004, 0.005]
BRACS	Aligned	Severe	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Severe	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Severe	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Severe	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe	CFAL	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Aligned	Severe	CFAL	Peak memory (bytes)	1.138e+09	[1.138e+09, 1.138e+09]
BRACS	Aligned	Severe	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Severe	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Peak memory (bytes)	512	[512, 512]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Severe	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Aligned	Severe	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Severe	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Severe	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Severe	Focal loss	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	OKO	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	OKO	Peak memory (bytes)	3.261e+07	[3.261e+07, 3.261e+07]
BRACS	Aligned	Severe	OKO	Processed examples	2983	[2983, 2983]
BRACS	Aligned	Severe	OKO	Parameters	3591	[3591, 3591]
BRACS	Aligned	Severe	OKO	Unique exposures	-8.933	[-10.47, -7.467]
BRACS	Aligned	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Severe	Diversity	Accelerator hours	0	[0, 0.001]
BRACS	Aligned	Severe	Diversity	Peak memory (bytes)	8.334e+05	[8.238e+05, 8.43e+05]
BRACS	Aligned	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Aligned	Severe (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Aligned	Severe (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Aligned	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	cRT	Accelerator hours	0.001	[0.001, 0.002]
BRACS	Aligned	Severe (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Aligned	Severe (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Severe (spread)	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	Focal loss	Processed examples	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Severe (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Aligned	Severe (spread)	OKO	Peak memory (bytes)	-9.001e+05	[-9.001e+05, -9.001e+05]
BRACS	Aligned	Severe (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Aligned	Severe (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Aligned	Severe (spread)	OKO	Unique exposures	-26.4	[-28.47, -24.13]
BRACS	Aligned	Severe (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Difficulty	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Aligned	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Aligned	Severe (spread)	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe (spread)	Diversity	Peak memory (bytes)	1.155e+06	[1.07e+06, 1.241e+06]
BRACS	Aligned	Severe (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Accelerator hours	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Balanced (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Reversed	Balanced (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Reversed	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Balanced (spread)	cRT	Peak memory (bytes)	1.319e+07	[1.319e+07, 1.319e+07]
BRACS	Reversed	Balanced (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Balanced (spread)	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Balanced (spread)	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Reversed	Balanced (spread)	OKO	Peak memory (bytes)	-6.161e+05	[-6.161e+05, -6.161e+05]
BRACS	Reversed	Balanced (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Reversed	Balanced (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Balanced (spread)	Diversity	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Diversity	Peak memory (bytes)	5.248e+07	[5.225e+07, 5.274e+07]
BRACS	Reversed	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
BRACS	Reversed	Moderate	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
BRACS	Reversed	Moderate	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Moderate	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Moderate	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
BRACS	Reversed	Moderate	CFAL	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Moderate	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Reversed	Moderate	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Moderate	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Moderate	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
BRACS	Reversed	Moderate	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Moderate	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Focal loss	Accelerator hours	0	[0, 0.001]
BRACS	Reversed	Moderate	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Focal loss	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Reversed	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Reversed	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	OKO	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	OKO	Peak memory (bytes)	-3.54e+05	[-3.54e+05, -3.54e+05]
BRACS	Reversed	Moderate	OKO	Processed examples	2983	[2983, 2983]
BRACS	Reversed	Moderate	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Moderate	OKO	Unique exposures	-0.067	[-0.2, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Moderate	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Moderate	Diversity	Peak memory (bytes)	-9.537e+06	[-9.563e+06, -9.516e+06]
BRACS	Reversed	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Severe	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Severe	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Severe	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Severe	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe	CFAL	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Reversed	Severe	CFAL	Peak memory (bytes)	1.138e+09	[1.138e+09, 1.138e+09]
BRACS	Reversed	Severe	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Severe	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Nominal support	Processed examples	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Severe	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Reversed	Severe	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Severe	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Severe	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Severe	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	OKO	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	OKO	Peak memory (bytes)	3.261e+07	[3.261e+07, 3.261e+07]
BRACS	Reversed	Severe	OKO	Processed examples	2983	[2983, 2983]
BRACS	Reversed	Severe	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Severe	OKO	Unique exposures	-16.2	[-17.8, -14.67]
BRACS	Reversed	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Severe	Diversity	Accelerator hours	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Severe	Diversity	Peak memory (bytes)	1.152e+06	[1.15e+06, 1.153e+06]
BRACS	Reversed	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Reversed	Severe (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Reversed	Severe (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Reversed	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	cRT	Accelerator hours	0.001	[0.001, 0.002]
BRACS	Reversed	Severe (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Reversed	Severe (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Severe (spread)	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Focal loss	Parameters	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Reversed	Severe (spread)	OKO	Peak memory (bytes)	-9.001e+05	[-9.001e+05, -9.001e+05]
BRACS	Reversed	Severe (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Reversed	Severe (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Severe (spread)	OKO	Unique exposures	-79.53	[-84.07, -74.87]
BRACS	Reversed	Severe (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Difficulty	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Reversed	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Severe (spread)	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe (spread)	Diversity	Peak memory (bytes)	1.129e+06	[1.119e+06, 1.137e+06]
BRACS	Reversed	Severe (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Native	Balanced (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Native	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Balanced (spread)	cRT	Peak memory (bytes)	1.319e+07	[1.319e+07, 1.319e+07]
BRACS	Native	Balanced (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Balanced (spread)	cRT	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Balanced (spread)	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Native	Balanced (spread)	OKO	Peak memory (bytes)	-6.161e+05	[-6.161e+05, -6.161e+05]
BRACS	Native	Balanced (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Native	Balanced (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Balanced (spread)	Diversity	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Diversity	Peak memory (bytes)	5.248e+07	[5.225e+07, 5.274e+07]
BRACS	Native	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Moderate	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Moderate	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	CE + soft MCC	Processed examples	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Moderate	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	CFAL	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Moderate	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Native	Moderate	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Moderate	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Moderate	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Moderate	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
BRACS	Native	Moderate	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Moderate	cRT	Parameters	0	[0, 0]
BRACS	Native	Moderate	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Focal loss	Accelerator hours	0.001	[0, 0.001]
BRACS	Native	Moderate	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Moderate	Focal loss	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Independent support	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Native	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Native	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Native	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	OKO	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	OKO	Peak memory (bytes)	-3.54e+05	[-3.54e+05, -3.54e+05]
BRACS	Native	Moderate	OKO	Processed examples	2983	[2983, 2983]
BRACS	Native	Moderate	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Moderate	OKO	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	Difficulty	Accelerator hours	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Moderate	Diversity	Peak memory (bytes)	-7.995e+06	[-7.997e+06, -7.991e+06]
BRACS	Native	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Native	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Severe	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Severe	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Severe	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Severe	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Native	Severe	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Severe	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe	CFAL	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Native	Severe	CFAL	Peak memory (bytes)	1.138e+09	[1.138e+09, 1.138e+09]
BRACS	Native	Severe	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Severe	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Severe	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Severe	Nominal support	Parameters	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Severe	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Native	Severe	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Severe	cRT	Parameters	0	[0, 0]
BRACS	Native	Severe	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Severe	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Severe	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Severe	Focal loss	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Native	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Severe	OKO	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	OKO	Peak memory (bytes)	3.261e+07	[3.261e+07, 3.261e+07]
BRACS	Native	Severe	OKO	Processed examples	2983	[2983, 2983]
BRACS	Native	Severe	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Severe	OKO	Unique exposures	-14.47	[-16.27, -12.6]
BRACS	Native	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Severe	Diversity	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Diversity	Peak memory (bytes)	1.152e+06	[1.15e+06, 1.153e+06]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Native	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Prevalence	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Severe (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Native	Severe (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Native	Severe (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Native	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	cRT	Accelerator hours	0.001	[0.001, 0.002]
BRACS	Native	Severe (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Native	Severe (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Severe (spread)	cRT	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Severe (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Native	Severe (spread)	OKO	Peak memory (bytes)	-9.001e+05	[-9.001e+05, -9.001e+05]
BRACS	Native	Severe (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Native	Severe (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Severe (spread)	OKO	Unique exposures	-70.87	[-75.87, -65.8]
BRACS	Native	Severe (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Difficulty	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Native	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Severe (spread)	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe (spread)	Diversity	Peak memory (bytes)	1.129e+06	[1.119e+06, 1.137e+06]
BRACS	Native	Severe (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	—	Balanced	Balanced sampling	Parameters	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Accelerator hours	0.014	[0.014, 0.014]
BRACS	—	Balanced	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Parameters	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	—	Balanced	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	—	Balanced	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Parameters	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	—	Balanced	CFAL	Accelerator hours	0.004	[0.004, 0.004]
BRACS	—	Balanced	CFAL	Peak memory (bytes)	1.121e+09	[1.121e+09, 1.121e+09]
BRACS	—	Balanced	CFAL	Processed examples	0	[0, 0]
BRACS	—	Balanced	CFAL	Parameters	-7	[-7, -7]
BRACS	—	Balanced	CFAL	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Nominal support	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	—	Balanced	Nominal support	Processed examples	0	[0, 0]
BRACS	—	Balanced	Nominal support	Parameters	0	[0, 0]
BRACS	—	Balanced	Nominal support	Unique exposures	0	[0, 0]
BRACS	—	Balanced	cRT	Accelerator hours	0.002	[0.002, 0.002]
BRACS	—	Balanced	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	—	Balanced	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	—	Balanced	cRT	Parameters	0	[0, 0]
BRACS	—	Balanced	cRT	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	—	Balanced	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	Focal loss	Processed examples	0	[0, 0]
BRACS	—	Balanced	Focal loss	Parameters	0	[0, 0]
BRACS	—	Balanced	Focal loss	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Independent support	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Independent support	Processed examples	0	[0, 0]
BRACS	—	Balanced	Independent support	Parameters	0	[0, 0]
BRACS	—	Balanced	Independent support	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Accelerator hours	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	—	Balanced	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	—	Balanced	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	—	Balanced	OKO	Peak memory (bytes)	-6.161e+05	[-6.161e+05, -6.161e+05]
BRACS	—	Balanced	OKO	Processed examples	2727	[2727, 2727]
BRACS	—	Balanced	OKO	Parameters	3591	[3591, 3591]
BRACS	—	Balanced	OKO	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Difficulty	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Difficulty	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	—	Balanced	Difficulty	Processed examples	0	[0, 0]
BRACS	—	Balanced	Difficulty	Parameters	0	[0, 0]
BRACS	—	Balanced	Difficulty	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	—	Balanced	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	—	Balanced	Diversity	Peak memory (bytes)	1.817e+07	[1.744e+07, 1.883e+07]
BRACS	—	Balanced	Diversity	Processed examples	0	[0, 0]
BRACS	—	Balanced	Diversity	Parameters	0	[0, 0]
BRACS	—	Balanced	Diversity	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Prevalence	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Prevalence	Processed examples	0	[0, 0]
BRACS	—	Balanced	Prevalence	Parameters	0	[0, 0]
BRACS	—	Balanced	Prevalence	Unique exposures	0	[0, 0]

Table 31: Matched computational-cost comparisons against cross-entropy in the same condition, for every roster method and every recorded cost endpoint, with crossed patient-bootstrap intervals.

The exposure budget is matched to within 0.6% across the roster, which is the property that makes the recovery comparison a comparison of methods rather than of data volume. Every method processes the same 503,880 examples as cross-entropy to within rounding, except set learning, which exceeds it by 2,727 to 2,983 examples because its per-update example count depends on a tuned set size and the derived update count must be integral, and classifier retraining, which exceeds it by 129. Post-hoc logit adjustment trains nothing of its own and reuses the cross-entropy checkpoint, so its processed-example difference is the full budget with a negative sign; its cost is a single pass to fit one scalar.

That matched exposure is achieved through unequal update counts, which is the intended trade. Set learning takes roughly 3,800 fewer updates than cross-entropy at 641 more examples per update, and classifier retraining splits its budget between two stages in a fixed ratio. Accelerator time, peak memory, and parameter counts differ by amounts their intervals absorb in every case, so no roster method’s recovery on this dataset is bought with materially more computation than the reference.

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